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Graduation Project - ENCS5300

Towards Smart Farming: An IoT-Based Optimized Agriculture System

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Section: 43

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Abstract

Agriculture in Palestine is considered the most significant sector that contributes to the gross domestic product. It is the pillar of the Palestinian economy and central to Palestinian identity. Reviewing the recent reports by international organizations and the Ministry of agriculture show the huge destruction in this sector. Recent reports by the United Nations show that less than 50% of households in Palestine were food secure for the last three years smart agriculture is introduced as a solution to improve the performance of the crop production process and secure the food sector. This project aims to adapt the Internet of Things and new technologies to improve the production of the agriculture sector in Palestine. The outcomes of this project will save the farmer time and minimize the effort required in the production process.

المستخلص

الاقتصاد الفلسطيني ومحور الهوية الفلسطينية، وبمراجعة التقارير الأخيرة الصادرة عن المنظمات الدولية ووزارة الزراعة يتبين حجم الدمار الذي لحق بهذا القطاع، حيث أظهرت التقارير الأخيرة الصادرة عن الأمم المتحدة أن أقل من ٥٠ % من الأسر في فلسطين كانت آمنة غذائياً خلال السنوات الثلاث الماضية، ويتم طرح الزراعة الذكية كحل لتحسين أداء عملية إنتاج المحاصيل وتأمين قطاع الغذاء، ويهدف هذا المشروع إلى تكييف إنترنت الأشياء والتقنيات الحديثة لتحسين إنتاج قطاع الزراعة في فلسطين، وستوفر نتائج هذا المشروع الوقت للمزارع وتقليل الجهد المطلوب في عملية الإنتاج.

List of Abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
CV	Computer Vision
DL	Deep Learning
IoT	Internet Of Things
ML	Machine Learning
VGG	Visual Geometry Group
YOLO	You Only Look Once
FC	Fully Connected
FPS	Frames Per Second
mAP	Mean Average Precision
IoU	Intersection Over Union
CS	Confidence Score
NMS	Non-Maximum Suppression
EC	Electrical Conductivity
LMV	Lettuce Mosaic Virus
WPAN	Wireless Personal Area Network
LDR	Light Dependent Resistor
WLAN	Wireless Local Area Network
LoRa	Low-Rank Adaption
LoRaWAN	Long Range Wide Area Network
DHT11	Temperature And Humidity Sensor

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Chapter 1

Introduction

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1.1 Overview

In Palestine, the cultivation of tomatoes, cucumbers, and lettuce is played a vital role in the agricultural sector, providing significant economic benefits and creating employment opportunities. However, numerous challenges are faced by farmers, including pest and disease management, resource constraints, and restrictions imposed by the occupation. These issues are aimed to be addressed by this project through the exploration of sustainable farming approaches that utilize edge computing and the Internet of Things (IoT). IoT is a network of interconnected physical sensors that collect and transfer data through the internet.

Additionally, the challenges of plant care for farmers through the proposal of a system that monitors essential factors such as humidity, light intensity, and temperature. Real-time data on soil and environmental conditions are collected using IoT sensors. Furthermore, plant diseases will be detected and managed, ultimately improving crop care and productivity, with the assistance of advanced machine learning (ML) algorithms integrated with Computer Vision (CV) systems. Sustainable farming practices will be promoted and crop welfare enhanced for Palestinian farmers by implementing IoT solutions for both agriculture and houseplant care.

1.2 Motivation

Concerns about food security have been raised by population growth and rapid urbanization, as traditional farming methods may be insufficient to meet rising food demand. Climate variability further compound agricultural challenges, including unpredictable weather and water scarcity. By 2030, water demand is expected to exceed supply by 40%, while 20% of agricultural land may be damaged. Despite pesticide use, pathogens continue to evolve, posing a major challenge to both the quantity and quality of crops. Recent studies have indicated that 42% of key global food crops are affected by agricultural pests. Additionally, significant economic losses are caused by the continuous emergence of new pests and fungi, making it impossible for farmers to identify these pests to control them, which can exacerbate crop losses that may lead to famine [**?**].

To address these pressing issues, integrating digital technologies such as Artificial Intelligence (AI) and IoT within the framework of Agriculture 4.0 is vital. Early detection of plant diseases is critical to boost crop production, optimize resource use (e.g., water and energy), and increase sustainability. This enables accurate environmental monitoring, predictive analytics, and effective farm management [**?**].

The idea starts with providing farmers a smart system that detects diseases in plant leaves, helping them monitor and understand their crops' health more effectively. The system works by allowing the user to send an image of a plant leaf to be classified (healthy or diseased) accurately, with the name of the detected disease displayed. In addition to image analysis, the system measures key environmental and soil factors, including light intensity (LDR), air humidity, air temperature, soil humidity, soil temperature, electrical conductivity (EC), and pH. A warning will be sent if one of the percentages is not at the normal level, and if this percentage is one of the causes of this disease. It also provides helpful tips to guide the farmer in restoring conditions to the optimal range. This allows for quick action to protect the plant and prevent further damage.

1.3 Problem Statement

Many challenges are faced by Palestinian farmers due to the Israeli occupation. Restricted access to land, water, and essential agricultural resources is suffered by them. Decreased productivity, poor agricultural practices, and limited use of modern agricultural technologies are brought about by these restrictions, coupled with population growth and increased food consumption. Additionally, they struggle with water shortages, soil degradation, and pest issues, further complicating agricultural efforts. The urgent need for new strategies to promote sustainable development and ensure food security is underscored.

The use of smart farming technologies in cultivating tomatoes, cucumbers, and lettuce is being investigated in this study. Leaf diseases are detected, soil and environmental conditions are checked, and plant growth is optimized using wireless sensor networks and CV. A system that can help people with their plant care habits wherever they are is clearly needed. Farmers should be provided with accurate information on humidity levels, light intensity, temperature, and soil conditions, and help farmers detect leaf diseases by this system.

Major problems such as pest control, efficient irrigation, and nutrient enhancement are helped to be solved by these technologies. Specific solutions are provided for Palestinian farmers by the study using these technologies. This helps the farmers deal with the challenges caused by the occupation and adopt sustainable agricultural practices. Smart agricultural systems that ensure food security and resilience for future generations in Palestine are aimed to be created by this approach.

1.4 Aims and Objectives

1.4.1 Aims

- Farmers, especially in Palestine, are helped to improve the performance of tomato, lettuce, and cucumber crops and increase their quality and productivity on the Palestinian markets. This is achieved through the advancement of agricultural methods and the identification of plant diseases.
- Plant diseases are detected early and the type of disease is determined as accurately as possible to help the farmer treat plant diseases.
- The concept of IoT in agriculture is used to make agriculture smart, as AI and sensors are combined to obtain high performance and precision regarding plant diseases.
- Plant disease is determined with high accuracy by using AI in a way that serves the farmer, as images of plant leaves are obtained, analyzed.
- While soil temperature, humidity, EC and PH, air temperature and humidity and light intensity are detected by the sensors and provided to the farmer so that required percentages for each plant can be determined and the farmer can be explained if one of the causes of the disease is the sensor reading and appropriate advice can be given to him.
- Palestinian farmers use Telegram Bot to tracking and monitoring many parts of their farms and stages of plant growth remotely via phone.

1.4.2 Objectives

- To study the related diseases of tomatoes, lettuce, and cucumbers, ensuring accurate identification of these diseases. To improve thier performance and increase their quality and productivity.
- To investigate the parameters that may affect crop production and their impact on plant health and yield.
- To improve and enhance production performance by adjusting the best environmental conditions and using advanced agricultural methods.

1.5 Report Organization

- In the first chapter, a clear overview of the project is provided, the factors that motivated us to undertake it are discussed, and its objectives and goals are outlined.
- In the second chapter, which presents previous studies about the project, many AI concepts and technologies were discussed; the three plants on which the project is based (tomatoes, lettuce, and cucumbers) were also presented with their sensors values, and the diseases of each plant were talked about.
- In the third chapter, which presents the methodology, a comparison between the communication system and the microcontrollers that could be used, the system requirement, the sensors used in the project, and the mechanism of linking them with the microcontroller, also the reason of using the Telegram Bot as a communication tool was mentioned as well as the system architecture.
- In the fourth chapter, which demonstrates the report result, starting with the dataset that have been chosen for each plant with the AI result of training this data with a comparison between the report training result and similar work, and the final result shown in Telegram Bot the disease detection with the sensor results.
- In the fifth chapter, which concludes the report and presents the future work.

1.6 Flow Chart

This is the flowchart for the report steps, shown in Figure 1.1.

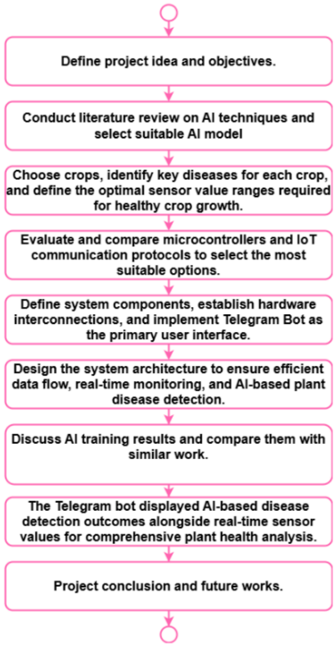

Figure 1.1: The flowchart.

Chapter 2

Literature Review

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2.1 Overview

This chapter provides an extensive review of research regarding the most suitable AI models and their techniques that can be used to detect leaf disease in tomato, lettuce, and cucumber crops.

The selected model is explained, along with the reasons for its choice. The reasons for choosing these three crops are explained. In addition, the most common plant diseases for the three crops will be discussed. The goal is to use this chosen AI model and crops in our leaf disease detection system to give useful information that can help farmers manage their crops better and meet their farming needs more easily.

2.2 Concepts and Techniques

2.2.1 Object Detection

To detect pests and diseases in leaves and plants, many solutions have been proposed, including CV which analyzes the leaf image and looks for signs of disease infestation. In addition, many systems have been developed to collect environmental information to help detect pests and diseases. Some systems combine various technologies to solve this problem [?]. Object detection and classification are important among different CV tasks that are applied in applications such as facial recognition, object tracking, autonomous driving, object character recognition, etc. With improved resources, better results can be obtained by training the computer to detect and classify objects [?].

Object detection combines two concepts: image localization and classification. When an image is provided as input, the model processes it to identify and locate multiple objects within the scene. The output consists of the 'x', 'y" coordinates along with the height and width parameters of each detected object, forming bounding boxes around objects of interest. In addition, each bounding box is assigned a corresponding class label indicating the type of object detected [?]. Figure 2.1 shows an example of leaf object detection.

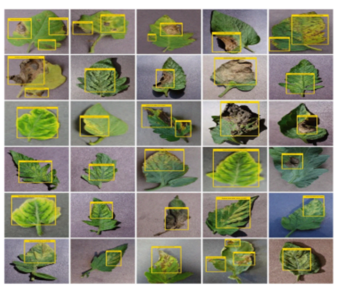


Figure 2.1: Example of leaf object detection.

Image classification is a common technique in CV that assigns the object in an image to one of the predefined class categories. The output can be one class or multiple class labels. Figure 2.2

displays pictures of the healthy and sick leaves of various plants.

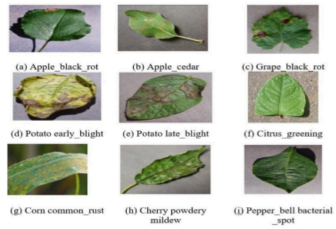


Figure 2.2: Healthy and diseased leaf images of the different plants.

Image localization process involves finding the particular position of an object within an image or video, then the output will have coordinates for x and y that define the object's location, allowing for the creation of bounding boxes around it. The localization for object detection is shown in Figure 2.3.

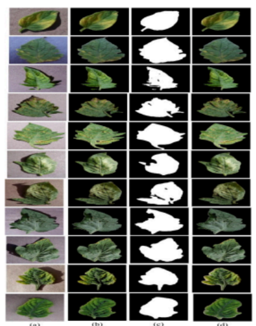


Figure 2.3: Example of image localization.

When Image localization is combined with image classification, this offers both locates objects in an image and categorizes them into different classes. Although when multiple objects are present in an image, localization methods alone are not enough; but the object detection method could be used to identify and classify all objects [**?**].

2.2.2 Neural Networks

A neural network consists of layers of neurons, which can be categorized into visible and hidden layers as shown in Figure 2.4. The input layer is where data is fed into the deep learning (DL) model, which is a subfield of ML based on multilayered artificial neural networks that are inspired by the human brain, while the output layer provides the final prediction. To produce a prediction, various computations involving thresholds and weights occur within the hidden layers. Forward propagation describes the movement of data from the input layer to the output layer, whereas backpropagation is the reverse process that adjusts the weights to reduce prediction errors.

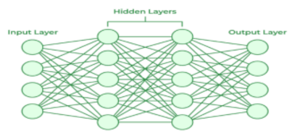


Figure 2.4: Neural network layers.

In IoT-based smart farming, neural networks are important because they provide advanced solutions for precision agriculture through various applications such as disease detection, soil condition monitoring, and optimizing irrigation systems. By leveraging sensors to gather data, these systems process information using ML algorithms to provide actionable and applicable insights. This integrated strategy enables farmers to make educated decisions, decrease resource waste, and increase yields [?].

2.2.3 Convolutional Neural Network (CNN)

The CNNs are type of deep neural networks. In the CV domain, it is designed for image classification, object detection, recognition of facial and handwritten characters, voice, text, and video processing. These networks are effective in detecting patterns in images, such as identifying plant diseases, because they automatically extract components such as edges and textures from raw image data. Also, it uses a hierarchical processing structure, so the initial layer detects simple features and the later layers recognize more complex patterns. CNN consists of layers as shown in Figure 2.5: the input layer (e.g., image data), pooling layers for downsampling, convolutional layers for feature extraction, and fully connected (FC) layers for classification leading to the output layer [?].

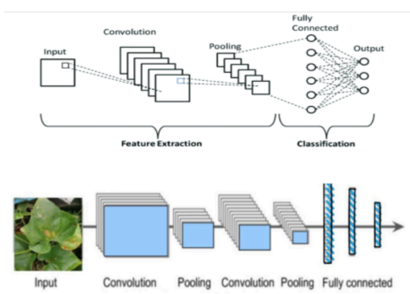


Figure 2.5: Architecture of a CNN.

As shown in the Figure 2.5, in the first the input image goes to the convolution layer; it's the most important part because it stores all unique properties in the image by looking at tiny parts (known as filters or kernels) in the image to try to detect basic features like colors, edges, and textures, and slide over the image to detect patterns in all image sections while allowing to determine the quantity of data that must be processed at one time. All of these help the network understand what is in the image. After that, pool the layers to reduce the image size and take only important information. Then the last step, a FC layer, combined all features and used them to make the final decision [?].

For example, EfficientNetB0 is a model based on CNN that has powerful performance in plant disease detection and achieves high accuracy with a value of 99.69 in detecting different diseases from datasets like Plant Village. CNN also achieves high accuracy, as shown in Table 2.1 [?].

Table 2.1: Accuracy for CNN and EfficientNetB0.

Model	**Accuracy (%)**	**Precision (%)**	**Recall (%)**
CNN	96.44	94.22	94.22
EfficientNetB0	99.69	98.27	98.26

2.2.4 You Only Look Once (YOLO)

Which is an algorithm for real-time object detection designed for single-step processing. During training and testing, YOLO views the entire image, so it utilizes the full image's features to execute both object detection and localization for each bounding box. In YOLO, object detection is redefined as a regression problem, so it immediately predicts class probabilities and bounding boxes from full images in a single evaluation. Thus, YOLO predicts the bounding boxes of all object classes in an image simultaneously in one pass using a single neural network, making it faster and more effective [?].

One-stage detectors like YOLO make object classification and detection in a single step, achieving faster inference rates essential for real-time applications. For example, YOLOv5 can reach detection speeds over 100 Frames Per Second (FPS). In contrast, two-stage detectors like Faster Region-Based Convolutional Neural Networks offer about 10 FPS. YOLO is faster and more accurate than traditional models because it uses all image properties to make predictions, enabling efficient global reasoning and end-to-end learning. It is suitable for applications like surveillance and autonomous driving. This model enables accurate and highly effective early detection, identifying and localizing disease symptoms on plants, which is essential to maintain food security and minimize crop losses.

This model has many versions, as shown in Figure 2.6. It has greatly improved in speed and accuracy since its first release. Introduced in 2015, it improved extremely in speed and accuracy over time. The original YOLO (YOLOv1) processed images at 45 FPS in real time, while faster YOLO processed images at 155 FPS with double the mean average precision (mAP). Later versions, such as YOLOv11, are now faster than 290 FPS. YOLO versions from YOLOv5 to YOLOv11 have shown constant improvement, focusing on speed, accuracy, and handling several classes through novel loss functions, advanced training methods, and architectural modifications [?].

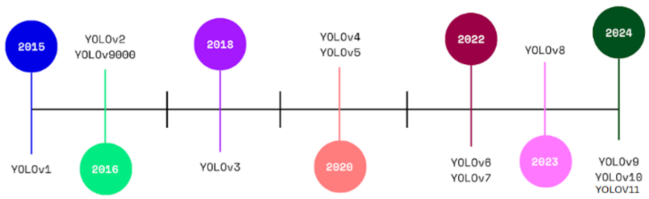


Figure 2.6: Versions of YOLO model.

The YOLO model performs object detection by first taking an input image and resizing it to a standard size, then dividing the image into an N×N grid of cells. Each grid cell is responsible for detecting objects whose centers fall within it. A single CNN is applied to the entire image, and each cell predicts objects based on whether their centers lie inside that cell. These steps allow YOLO to process the image in parallel and in one pass. Each grid cell predict a fixed number of bounding boxes (e.g., three), assigns a confidence score (CS) to each box, and outputs class probabilities such as 'person' or 'car' for the objects within those boxes. As a result, YOLO works on the full image and on all objects in this image. Finally, detection results are filtered based on CS, allowing YOLO to perform localization and classification simultaneously over the full image.

The bounding box CS in object detection shows how confident the model is that there is an object inside the box and how accurate the prediction is. In other words, it answers two questions: (1) What type of object is inside the box? and (2) How does the prediction box fit the object?

The CS equals zero if no object is present in the cell; otherwise, it equals the intersection over union (IoU) between the ground truth and the predicted box.

To calculate the CS for each bounding box, use the equation:

$$\Pr(\text{Class}_i \mid \text{Object}) \times \Pr(\text{Object}) \times \text{IoU} = \Pr(\text{Class}_i) \times \text{IoU} \tag{2.1}$$

Where the CS is given by:

$$\text{CS} = \Pr(\text{Object}) \times \text{IoU}_{\text{truth-pred}} \tag{2.2}$$

$Pr(Class_i \mid Object)$ is the class probability that there is an object. IoU measures how well the predicted box matches the actual object. A higher IoU means a better fit. Figure 2.7 highlights the IoU equation, and Figure 2.8 shows the IoU values in different situations. As shown, performance improves with increasing IoU value [?].

Figure 2.7: IoU of two boxes.

Figure 2.8: IoU values in different box positions.

As simple as we talk, YOLO divides the image into a grid, and each grid cell makes a prediction: 1. Is there an object in the cell? 2. What class is this object? But it doesn't make these two predictions at the same time. If the first prediction answer is yes, this means it doesn't contain an object in the box, so after the result, yes, the grid class can make a second prediction [?], as shown in Figure 2.9.

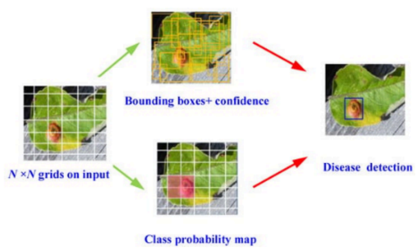


Figure 2.9: YOLO object detection on plant leaves.

Five parameters describe a bounding box:

- **Center coordinates (x, y):** show the center of the box relative to the grid cell's boundaries.
- **Width (w):** The width of the bounding box.
- **Height (h):** The height of the bounding box.
- **CS:** A probability (0–1) indicating the likelihood of an object in the bounding box.
- **Class prediction:** The class label for detected objects such as people, cars, dogs, etc.

After all predictions are made, Non-Maximum Suppression (NMS) removes redundant boxes that predict the same object. This technique eliminates boxes with high overlap with higher-confidence boxes to keep only the most accurate box for each object.

NMS works by: Sorting predicted boxes by CS value, then selecting the box with the highest CS. After that, removing any boxes overlapping it if the IoU is greater than a threshold (e.g., 0.5) [?].

2.2.5 Visual Geometry Group (VGG)

It is a deep CNN; It has a straightforward sequential structure of layers, which makes it easier to implement and adapt for specific tasks. It has two popular architectures, VGG16 it has 16 layers, (13 convolution, 2 FC and 1 SoftMax). Although VGG19 consists of 19 layers (16 convolution and 3 connected layers), which makes it higher depth, powerful and can capture difficult features in image classification [?].

VGG16 and VGG19 have the same simple architecture with (3×3) convolution layers. It is effective in detecting detailed features of an image, such as detecting damaged leaves on plants. And (2×2) MAX-pooling layers after every two or four convolution layers to reduce the dimensionality. The VGG19 allows for more complex features but makes the model heavier. VGG16 on the other hand is lighter, because it has lower convolution layers, which leads to lower convolution calculations and more efficiency. So, the VGG-16 model is better.

Table 2.2 shows a comparison between the VGG16 and VGG19 models. The VGG16 model shows better performance than VGG19 since it reaches 91.9% accuracy together with 89% precision, but VGG19 only achieves 88.2% accuracy with 74% precision. The F1 score indicates VGG16 better than VGG19 through its 70% score, although VGG19 demonstrates higher recall at 75% but lower precision reaches only 72% [?].

Table 2.2: Comparison Between Two VGG Models.

NO.	Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
1	VGG 16	91.9	89	72	70
2	VGG 19	88.2	74	75	69

2.2.6 Residual Networks (ResNet)

It is a common neural network design used for DL CV applications like image classification, object detection, and image segmentation. ResNet architecture in common includes many convolution layers in the range of 18-152, but can be extended to support thousands of convolution layers.

ResNet was generated to solve the challenge of training very deep neural networks, such as vanishing or exploding gradients. ResNet enables the network layer to learn "residual mappings" rather than directly learning the mappings from input to output, which means instead of each layer needing to learn a full transaction from input to output, it only needs to learn the difference between them. simplify the learning process. To do this operation, ResNet used "skip connections" that allow specific layers to skip over others, preserve important data, and prevent problems with extremely deep networks. This design makes training effective and accurate when it has hundreds or even thousands of layers, making it a crucial instrument for DL [?].

2.3 Similar Work

Table 2.3 summarizes similar previous studies, detailing their methods and datasets used, as well as the advantages and disadvantages of each study in terms of accuracy, to help us confirm the most suitable model.

Table 2.3: Summaries of Various Research Works and Methods.

Work/Research	**Method**	**Dataset**	**Pros**	**Cons**	**Accuracy**
Identifying Common Pest and Disease of Lettuce Plants [?].	CNN	Images of healthy, diseased, and pest-affected lettuce plants.	Good feature learning; effective on images.	Requires training time; not real-time; needs tuning.	95.72%
Enhanced Plant Disease Detection Using CV YOLOv11 [?].	YOLOv11	Tomato leaf dataset with 10 disease classes and 1 healthy class.	Fast inference; suitable for real-time use; better performance than earlier YOLO versions.	May require more computational resources.	Up to 99.35%
Plant Leaf Disease Detection Using Ensemble Learning and Explainable AI [?].	VGG	VGG16 & VGG19 ensemble on PlantVillage dataset.	Strong feature extraction; deep architecture.	Slow; memory intensive; not real-time.	Over 90%
A ResNet50-DPA model for tomato leaf disease identification [?].	ResNet50	Tomato leaf images (10 disease classes).	Captures multi-scale features; outperforms standard ResNet.	Increased model complexity.	99.28%

2.4 YOLO: Selected Model.

In our leaf disease detection project, we selected YOLOv11 instead of models like CNN, ResNet, and VGG due to its faster, more precise, and real-time detection of leaf diseases. Unlike traditional CNNs and VGG, which primarily focus on classification, YOLOv11 combines both detection and classification at the same time (in a single pass), allowing it to identify and localize diseases directly on the leaf image using bounding boxes. This integrated approach eliminates the need for additional preprocessing steps such as cropping, making the system faster and more efficient. Although models like ResNet and VGG are known for high accuracy, they are computationally heavier and slower and are more unfit for real-time applications in agricultural settings. In contrast, YOLOv11 is lightweight and optimized for efficient deployment on cloud servers or edge devices like the ESP32, requiring minimal computing power. Compared to previous versions, YOLOv11 also delivers improved accuracy, making it even more effective for plant disease detection. Furthermore, YOLOv11 supports custom-labeled datasets, making it highly adaptable for training on specific crop diseases. These advantages make YOLOv11 an ideal choice for fast, reliable, and scalable disease detection in smart farming environments.

2.5 Leaf Diseases of Selected Crops: Tomato, Lettuce, and Cucumber

Three crops—tomato, lettuce, and cucumber—were selected for our project to detect leaf diseases in Palestine due to their critical importance in the region's agriculture and high consumption in local markets. Farmers from Jenin, Hebron, and Tulkarm confirmed that these crops are widely consumed and exported because of their adaptability to climate. Additionally, the short life cycle of crops was compatible at the time of our project. These crops play a key role in supporting food security and farmers income on the farms.

Plant leaf diseases are caused by pathogens like bacteria, viruses, and fungi or by environmental and physiological factors. They often appear as spots, wilting, fungal growth, or discoloration, which leads to poor photosynthesis and reduces crop productivity. Thus, early detection is crucial to protect agricultural crops and support the development of the farming sector [**?**].

2.5.1 Tomato Crop

Tomatoes are one of the most popular crops in terms of production. They can be grown year-round in different seasons and ways, with a growth period between 90 and 150 days. They are consumed fresh in large quantities and processed in several other ways. Economically, they are a high-demand crop and are exported in large quantities due to their climate adaptability.

Figure 2.10 shows that fresh tomato production in Palestine reached 186.85 million kilos in 2022, with a market share of 0.10%. Also, it shows the production decreased by 18.26% from the previous year. This highlights challenges in the agricultural sector and emphasizes the importance of detecting and treating diseases to enhance production [?].

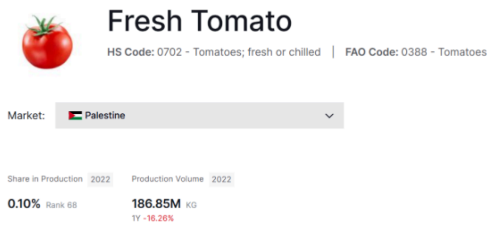


Figure 2.10: The production of fresh tomatoes in Palestine.

The chart in Figure 2.11 shows the percentage of tomato production in Palestine over the years, with a noticeable increase after the early 1990s and a peak in 2022 at 186,847.16 tonnes. This indicates the development of the agricultural sector and the importance of the tomato crop for both local consumption and export. This data is from the Food and Agriculture Organization [?].

Figure 2.11: Percentage of tomato production in Palestine over the years.

2.5.1.1 Tomato Leaf Diseases

1. **Early Blight (Alternaria solani):**

   Caused by the fungus Alternaria solani. It appears as large irregular spots and rings of black surrounded by bigger yellow regions of dead tissue. Leaf spots are characterized by concentric lines (like an oyster shell or bull's eye). Leads to a leaf drop. Occurs in warm, humid conditions that allow the fungus to grow. Figure 2.12 displays the leaf infection [?].

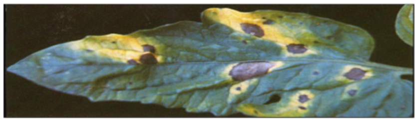


Figure 2.12: Early blight.

2. **Late Blight (Phytophthora infestans)**:

   Caused by Phytophthora infestans. Appears as dark brown, water-soaked sores on leaves; large irregular greenish-brown spots with a rough, greasy appearance. One of the most damaging tomato diseases, particularly in cold, wet conditions. Affects leaves and fruits. Figure 2.13 displays the leaf infection [**?**].

Figure 2.13: Late blight.

3. **Leaf Mold (Passalora fulva)**:

   Caused by the fungus Passalora fulva. Appears as green spots on the upper surfaces of older leaves with a moldy appearance on the underside. Occurs in high humidity and poor ventilation as the fungus spreads by wind currents, affecting tomato plants, especially in greenhouses. Figure 2.14 displays the leaf infection [**?**].

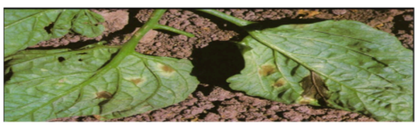


Figure 2.14: Leaf mold.

4. **Septoria Leaf Spot (Septoria lycopersici)**:

   Caused by the fungus Septoria lycopersici. Appears as circular, water-soaked lesions with gray centers, causing the leaves to die and fall off. Spreads rapidly in humid weather, as old crops and weeds contribute to formation and spread. Figure 2.15 displays the leaf infection [**?**].

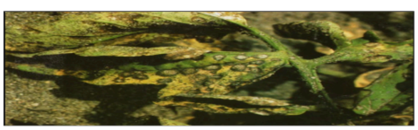


Figure 2.15: Septoria leaf spot.

5. **Bacterial Spot (Xanthomonas species)**:

   A bacterial infection causing dark brown, water-soaked spots that coalesce and turn black. Infected tissue falls off, leaving holes in the leaf, leading to leaf death. Spread by splashing water. Figure 2.16 displays the leaf infection [**?**].

Figure 2.16: Bacterial spot.

6. **Gray Leaf Spot (Stemphylium solani)**:

   A fungal disease caused by Stemphylium solani. Appears as small dark spots with gray centers, often on older lower leaves. Spots develop into larger necrotic areas that cause tissue to fall off, leaving a bullet-hole appearance. A golden halo may surround the dots. Yellowing, leaf drop, and leaf drop can occur in severe cases. Disease worsens when leaves are wet from rain or dew. Figure 2.17 displays the leaf infection [**?**].

Figure 2.17: Gray leaf spot.

2.5.2 Lettuce Crop

It is a leafy green vegetable from the Asteraceae family and is require in many Palestinian houses. The most famous type of lettuce in Palestine is green lettuce. It is a cool-season crop, so its grown best in spring and autumn outdoor, but it also can be grown year-round if it's grown in greenhouses or indoor where appropriate conditions are available and temperature are controlled.

Figure 2.18 shows lettuce production data in Palestine in 2022; it reached more than 1.42 million KG, ranking 76th globally with a share of 0.01% in global production. Also, shows a drop of 5.98% from the previous year. These results show the challenges of agriculture and the importance of improving agriculture and helping farmers understand their farm needs.

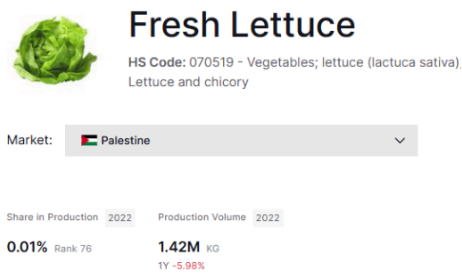


Figure 2.18: Lettuce in Palestine.

Figure 2.19 presents chart details about production volume over many years. It shows the worst year was 2010, with production around 500K KG, then its increase until reaching the best years of production was from 2014 to 2017, with production around 1.82M KG, then its increase until the last year recorded was 2022, with production of 1.42M [?].

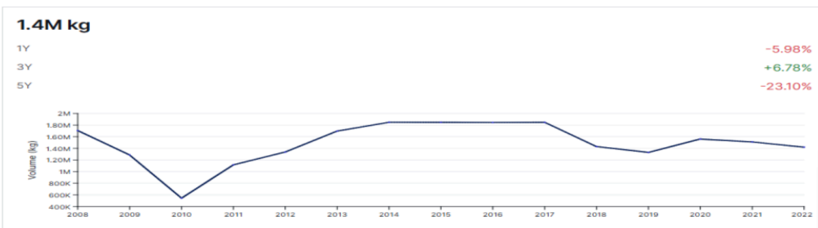
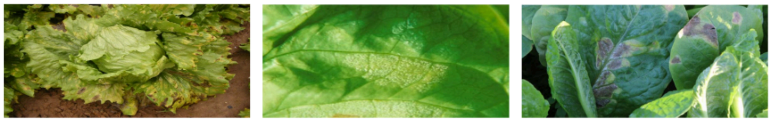


Figure 2.19: Production volume chart for lettuce in Palestine.

2.5.2.1 Leaf Diseases of Lettuce

1. **Downy mildew (Bremia lactucae):**

   The most common disease in lettuce, caused by the pathogen Bremia lactucae, Symptoms begin as pale-green or yellow angular patches. Then evaluation to white sporulation on the lower leaf in cool temperatures (15–20°C), as shown in Figure 2.20. The lesions then dry up and turn brown. Disease control includes managing environmental conditions to reduce humidity and leaf wetness. Rapid removal of infected plant residues after harvest reduces the spread of disease, especially in greenhouses [?].



(a) Outdoor crisphead lettuce plant infected by downy mildew.

(b) Sporulation of downy mildew on leaf underside.

(c) Downy mildew angular lesions that are chlorotic and necrotic.

Figure 2.20: Stages of the downy mildew disease on lettuce leaves.

2. **Septoria blight:**

   Caused by the fungus *Septoria lactucae*. The initial symptoms include small, irregular yellow spots appearing on the oldest leaves, as shown in Figure 2.21. These spots gradually enlarge, turn brown, dry out, and may eventually fall off, resulting in a ragged appearance of the leaves [**?**].

Figure 2.21: Septoria blight on lettuce.

3. **Fusarium wilt:**

   caused by Fusarium oxysporum f. sp. lactucae (FOL), the symptoms include wilting, yellowing, and browning of seedlings as shown in Figure 2.22, and stunting in older plants, often accompanied by a reddish-brown discoloration of the vascular system. Infection can occur at temperatures as low as 15°C. Control measures include avoiding planting in infested soil, using resistant or tolerant lettuce varieties (if available), and reducing soil movement between sites. For protected crops, soil disinfection through fumigation or hydroponic systems is recommended [**?**].

(a) Fusarium wilt that appears as wilting and yellowing of the leaves.

(b) Vascular discoloration of fusarium wilt.

Figure 2.22: Visual representation of fusarium wilt disease on lettuce.

4. **Lettuce mosaic virus (LMV):**

   Caused by a potyvirus. The symptoms differ based on the lettuce variety and timing of infection, causing stunted growth, yellowing lesions, and deformities, as shown in Figure

2.23 [?]. Effective management includes planting certified LMV-free seeds, weed control, aphid management, and use of resistant lettuce varieties [?].

Figure 2.23: Mosaic viruses appear as a mosaic of green and yellow colors on a diseased lettuce leaf.

5. **Bacterial leaf spot (Xanthomonas campestris):**

   Caused by Xanthomonas hortorum pv. Vitians (Xhv). The pathogen causes waterlogged spots that turn dark brown, as shown in Figure 2.24 [?], and these spots are exacerbated by cool, wet conditions, resulting in severe yield loss (up to 100%). Disease management focuses on using pathogen-free seed, avoiding overhead irrigation, crop rotation, removing debris, and applying copper hydroxide when detected [?].

Figure 2.24: Advanced signs of bacterial leaf spots.

2.5.3 Cucumber Crop

Cucumber is a commonly cultivated vegetable in Palestine, particularly in the West Bank and Gaza Strip, where it thrives in the Mediterranean climate. It is an important crop for both local consumption and commercial purposes. Palestinian farmers rely on cucumbers as a significant source of income, and this vegetable plays a vital role in the country's food security and agricultural economy. Figure 2.25 illustrates the production of fresh cucumbers in Palestine, which amounted to 112.08 million kilos in 2022, representing a market share of 0.12%. It also indicates that production decreased by 5.02% compared to the previous year. These indicate some challenges in the agricultural sector and emphasize the importance of identifying and addressing diseases to improve production [?].

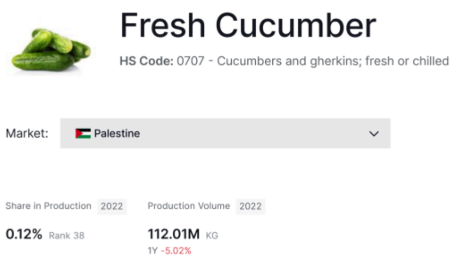


Figure 2.25: The production of fresh cucumber in Palestine.

The chart in Figure 2.26 illustrates the percentage of cucumber production in Palestine over the years. Notably, there was a significant increase in production after 2008, peaking in 2022 with a total of 112 million kilograms. This trend reflects the improvements and developments within the agricultural sector and underscores the importance of cucumbers, both for local consumption and export [?].

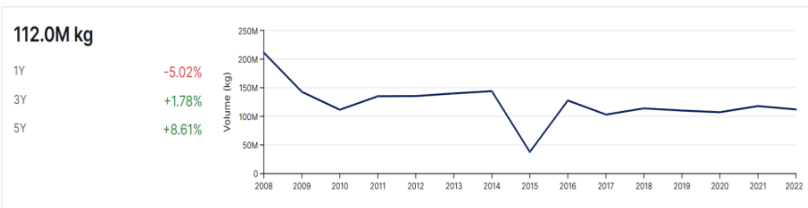


Figure 2.26: Percentage of cucumber production in Palestine in 2000s.

2.5.3.1 Leaf Diseases of Cucumber

1. **Anthracnose**

   One of the earliest symptoms of cucumber anthracnose, a common fungal disease that affects many plants, is yellow, water-soaked leaf spots. Warm, humid weather, especially with high humidity, is ideal for the disease to grow. Figure 2.27 illustrates how the illness appears on the leaves [?].

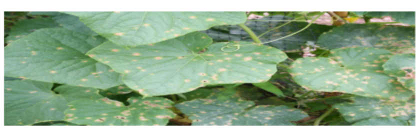


Figure 2.27: Anthracnose leaf disease for cucumber.

2. **Gummy stem blight**

The fungus "Mycosphaerella cucumis" causes brown lesions on the base and top of the stem of cucumbers. These lesions appear as golden gummy droplets, hence the name "gummy stem blight." The lesions appear as black spots, which are new spore-bearing fruiting bodies. Circular, elongated lesions may appear on the leaves as shown in Figure 2.28 [**?**].

Figure 2.28: Gummy stem blight leaf disease for cucumber.

3. **Bacterial wilt**

Cucumber beetles with stripes and spots are the vectors of the bacterial wilt disease. Early in the season, when plants are growing quickly, withering is more harmful, however symptoms can manifest at any time. The disease can quickly spread to the full plant, even though at first only a few vines may exhibit symptoms. Figure 2.29 illustrates how the illness appears on the leaves [**?**].

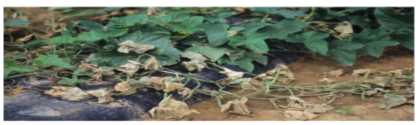


Figure 2.29: Bacterial wilt cucumber leaf disease.

4. **Downy mildew**

Downy mildew is a disease that mostly affects the leaves of cucumbers and similar crops, including pumpkins and melons. It is caused by the fungus-like water mold Pseudoperonospora cubensis, and it can quickly result in severe yield losses. Figure 2.30 illustrates how the illness appears on the leaves [**?**].

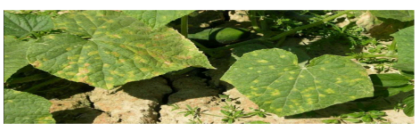


Figure 2.30: Downy mildew cucumber leaf disease.

2.5.4 Common Diseases Between Our Crops:

2.5.4.1 Powdery Mildew (Oidium species)

Powdery mildew is a common fungal disease affecting plants such as cucumbers, tomatoes, and lettuce. It appears as white or gray spots on the surfaces of leaves, stems, and fruits, often starting on older leaves with yellow spots. The leaves eventually die, remaining attached to the stem. In cucumbers, it can reduce photosynthesis and yield, while tomatoes may suffer from leaf curl and premature leaf drop. Lettuce may exhibit distorted growth and reduced quality. This disease thrives in warm, dry environments with high humidity and poor air circulation, making greenhouses more susceptible. Figure 2.31 shows the disease on the three plants [?].

Figure 2.31: Powdery mildew on cucumber, lettuce, and tomato.

2.6 Best Environmental Conditions for Lettuce, Tomato, and Cucumber Growth

In this section, the optimal environmental conditions for the growth of tomatoes, lettuce, and cucumbers are presented. Table 2.4 compares the ideal conditions required for each plant, such as temperature, humidity, light, and soil characteristics. Based on these values, the system helps the farmer determine whether there is a normal environmental condition around the plant or if there are any deficiencies or increases in one of the sensor readings so that the farmer can take action to address it. Moreover, the system assists in knowing whether such values of these sensors could be leading to the leaf disease of the plant. This reflects the variation in the needs of each crop based on its biological nature and optimal growing conditions.

Table 2.4: Optimal Growth Conditions for Lettuce, Tomato, and Cucumber.

Parameter	**Tomato**	**Lettuce**	**Cucumber**
Growth Period (days)	60–85 [?]	65–130 [?]	50-70 [?]
Air Temperature (° C) [?]	21-29	16-18	16-32
Air Humidity (%) [?]	60–75	50-70	70-85
Light (LDR Digital) [?]	High light	Moderate light	High light
Light (LDR Analog) [?]	100-500	500-1500	100-500
Soil pH [?]	5.5-7.5	6.0-7.0	5.5-7.5
Soil EC (dS/m) [?]	2.0 – 5.0	1.2–2.0	1.7-2.5
Soil Temperature (° C) [?]	21.1-35	4.4-26.7	15.6-35
Soil Humidity (%)	60–80 [?]	40-60 [?]	65–80 [?]

2.7 Conclusion

In conclusion, this chapter talked about different AI techniques and models like CNN, VGG, YOLO, and ResNet, and their application in DL for object detection with the importance of neural network components like convolution, pooling, and FC layers. In addition to that, we discuss some important disease detection in three plant leaves (tomato, lettuce, and cucumber) and why we chose these plants based on previous studies, along with the best environmental conditions with the growth period for each plant to improve agricultural techniques. These models and methods help in improving the accuracy and efficiency of plant disease detection.

Chapter 3

Methodology

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3.1 Introduction

This chapter analyzes IoT communication protocols: Bluetooth, Long Range Wide Area Network (LoRaWAN), and Wi-Fi, evaluating their suitability for different systems. It also provides a comparison of widely used microcontrollers to justify the hardware selection. The chapter determined and explained the core components of the system and explained how they are interconnected to enable efficient data collection and processing. Additionally, it highlights the role of the Telegram bot as an interactive communication tool and presents an overview of the system architecture, detailing how sensor data is transmitted to reach the results in the Telegram bot interface.

3.2 Comparison Between IoT Communication Protocols

Table 3.1 summarizes various communication protocols, which include detailed descriptions and representative uses of each protocol. LoRaWAN, Bluetooth, and Wi-Fi are data transfer protocols that work with ESP32 microcontrollers in IoT-based applications. We selected Wi-Fi as the preferred data transfer method using the ESP32 for sensor communication. Wi-Fi proved to be suitable for our system due to its excellent combination of operational range it covers 50–100 meters. Supports high data rates (up to hundreds of Mbps). Higher power usage and straightforward implementation capabilities for greenhouse and small farmland monitoring. LoRaWAN offers kilometer-level range, but this is unnecessary unless covering vast, remote areas. Extremely low data rate (0.3–50 kbps). Only good for basic data like temperature/humidity, low power, Requires additional transceivers and its need a Low-Rank Adaption (LoRa) gateway or private network setup. Although Bluetooth has a very short range (5–30 meters), its data rate is higher than LoRa, but still limited and unreliable for rich or frequent data. low power consumption.

Table 3.1: Comparison of IoT Communication Protocols.

Protocol	**Description**	**Use case**
LORAWAN	The low-power wireless communication system is designed for IoT devices to transmit data over long distances while consuming minimal power. It operates on unlicensed radio frequencies and utilizes spread spectrum modulation, enabling devices to communicate from distances of up to 20 kilometers. This system is particularly beneficial for rural areas or expansive regions where energy efficiency is a priority [?].	LoRaWAN aids agriculture by collecting real-time data from sensors. This information is sent to a central system to manage irrigation and detect pest problems early. Its ability to cover large areas with low power makes it ideal for managing farms, especially in remote areas [?].
BlueTooth	It has a broad range of 2.4 GHz applied in Wireless Personal Area Network (WPAN) using a star-bus topology, with a data rate of 1 Mbps, making it suitable for short-range communication (10–100 meters). It also has low power consumption between 1-35 mA [?].	Because it has low power and a short rang that makes it suitable for tracking the location of items, like tracking goods or medical equipments.

Protocol	Description	Use case
WIFI	WIFI is used in Wireless Local Area Network (WLAN), providing local area networks with fast internet access and operating in a star topology. It typically offers a data rate of 150 kbps and operates in the 2.4/5 GHz frequency bands. Its coverage range spans 10 to 50 meters, with relatively high power consumption. WiFi is well-known for its availability, flexibility, and cost-effectiveness. It enables connections between IoT devices, such as sensors and cameras, and servers to allow data transmission [?].	Allows data transmission for systems such as remote monitoring, home automation and environmental sensing systems by enabling local connectivity for smart devices in homes, workplaces, and farms.

3.3 Comparison Between Microcontrollers

Table 3.2 shows different types of microcontrollers that are widely used, focusing on their processor performance, memory capacity, and networking capabilities, along with cost breakdowns. The ESP32 emerges as the most suitable choice from the options provided because it contains dual-core processing capabilities along with 520 KB RAM and 16 MB flash memory storage while offering built-in Wi-Fi and Bluetooth, as well as high ADC and GPIO channel numbers. All these features make it the optimal choice for our project because it manages to find the perfect compromise between performance and cost while naturally linking to wireless technologies and providing robust capabilities for developing intricate IoT systems.

Table 3.2: Comparison of Popular Microcontrollers

Feature	Arduino Uno [?]	ESP8266 [?]	ESP32 [?]	Raspberry Pi Pico [?]	STM32F103 C8T6 [?]
Processor	8-bit AVR @ 16 MHz	Xtensa @ 80–160 MHz	Dual-core Xtensa @ 240 MHz	Dual-core Cortex-M0+ @ 133 MHz	Cortex-M3 @ 72 MHz
Flash Memory	32 KB	4 MB	4–16 MB	2 MB	64 KB
RAM	2 KB	80 KB	520 KB	264 KB	20 KB
GPIO Pins	14 (6 analog)	∼11	∼34	26	37
Wi-Fi	No	Yes	Yes	No	No
Bluetooth	No	No	Yes	No	No
ADC Channels	6 (10-bit)	1 (10-bit)	18 (12-bit)	3 (12-bit)	10 (12-bit)
Power Consumption	Low	Medium	Medium–High	Very Low	Low
USB Support	Yes (via ATmega16U2)	Yes (Micro-USB)	Yes (Micro/USB-C)	Yes (host/device)	Yes (via UART)
Estimated Cost	$5–10	$2–5	$4–10	$4	$2–4

3.4 Components of the System

3.4.1 ESP32 -WROOM-32D Microcontroller

The ESP32 shown in Figure 3.1 is essential for monitoring and managing a number of environmental factors in IoT-based smart farming and agriculture. It can gather information from all sensors connected with like soil and environmental sensors, then send it to a cloud platform or central system for real-time analysis. With a dual-core 240 MHz processor, 520 KB RAM, up to 16 MB flash memory, 18 × 12-bit ADC channels, and around 34 GPIO pins, the ESP32 offers excellent performance and flexibility. Its ease of integration and wide software support make it a reliable choice for scalable smart farming systems [?].

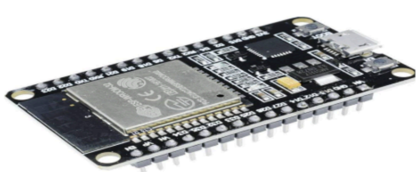
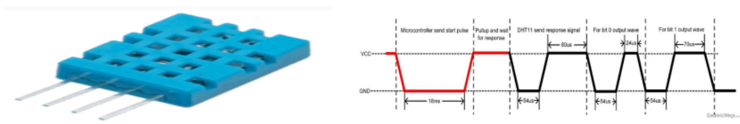


Figure 3.1: ESP32 Wi-Fi module and bluetooth.

3.4.2 Temperature and Humidity Sensor (DHT11)

The DHT11 sensor is a single-wire temperature and humidity sensor with a one-wire protocol. It is affordable, high-quality, and reliable, with a temperature range of 0°C to 55°C and humidity from 20 to 90%. It has four pins: one for VCC, which requires 3.3 to 5V DC power, and the other for the digital output pin. The sensor collects data from the environment and sends it to a microcontroller. The microcontroller sends a starting request for the sensor, which waits for a response. The sensor sends a pulse, and the communication starts. The data has five segments, each 8 bits long. The first two bits represent humidity as a decimal integer, while the subsequent two represent temperature in Celsius. The final section serves as a checksum, calculated by summing the humidity and temperature values. If the checksum does not align with the data, it signals an error in the received information. The sensor pin remains in a low-power state until the next start pulse [**?**]. Figure 3.2 shows the DHT11 sensor and its methology to send data.



(a) DHT11. **(b)** communication between DH11 and microcontroller.

Figure 3.2: DHT11 sensor and its communication with a microcontroller.

3.4.3 Light Dependent Resistor (LDR)

The LDRs shown in Figure 3.3 are compact sensors, so they are also called photoresistors, photoconductors, or photocells [**?**]. They are passive components whose resistance changes in response to light intensity—how much light has fallen on them. They convert light energy to electrical signals, and they are made up of a semiconductor substance that, when exposed to light, increases the amount of free charge carriers, lowering resistance. Because there are fewer free electrons available to carry current in the dark, the resistance is quite high [**?**]. It runs on +3.3V of voltage. Light is used as its input, and the output is an analog signal, and it uses LUX measuring units [**?**]. We will use this sensor to measure the light intensity falling on different types of plants. It measures the change in light intensity, which enables the system to determine whether the light intensity is suitable for the plant or not. In other words, is the plant getting enough light to support photosynthesis and promote healthy plant growth?

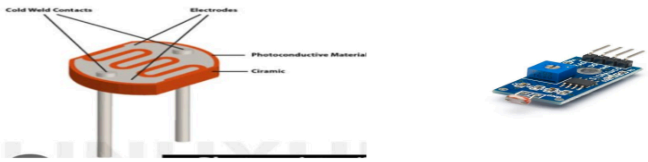


Figure 3.3: LDR.

3.4.4 RS485 Sensor (CWT-Soil-THCPH-S)

The sensor shown in Figure 3.4 is a 4-in-1 sensor that is used to measure soil temperature, humidity, EC, and PH. It's made of stainless steel, making it waterproof, rustproof, anticorrosive, long-term electrolysis resistant, and salt-alkali corrosion resistant. It is buried in the ground to send data. Its output is stable and has a fast response. It can also be used with different types of microcontrollers, including ESP32 because of its RS485 output, but it needs to connect with the LC-Electronics shown in Figure 3.5 to help the ESP32 to transmit the TTL-level RS-485 communication signal. Its power supply is between 5 and 30V.

To detect humidity, the sensor uses electromagnetic wave pulse technology. The sensor transmits electromagnetic pulses to the soil via its probe. The sent pulses interact with the soil dielectric properties that change because of moisture amounts. Water shows a greater dielectric constant than dry soil substances. When the soil moisture level alters, the electromagnetic pulses experience changes in behavior because of the shifting dielectric constant. The sensor uses detected changes to determine the soil's water content from the measurements [**?**]. Its humidity measuring range is between 0-100%, and its humidity resolution is 0.1%, and when the accuracy is between 0-50%, the sensor's readings can vary by up to ±2%, while when it's between 50-100%, it's a little bit lower; the variation is up to ±3%.

Although the temperature measuring range is approximately between -40 °C +80 °C, and its resolution is 0.1 °C while the accuracy is ±0.5 °C [**?**] the sensor uses a high-precision thermistor that is built into its stainless steel. Thermistors are heat-sensitive resistors whose resistance changes predictably with temperature changes. By monitoring these changes, the sensor accurately determines the soil temperature [**?**].

EC is affected by variables including salinity levels and moisture content and is calculated using low-frequency pulses. Changes in the soil's conductivity, which are brought on by dissolved ions, are found and utilized to determine the salinity of the soil [**?**]. Its measuring range is between 0–20000 µS/cm and its resolution is 1 µS/cm. If the EC is between 0–10000 µS/cm, the accuracy is ±3% full scale, while if the EC is between 10000–20000 µS/cm the accuracy is ±5% full scale [**?**].

Finally, the PH is used to measure acidity or alkalinity in the plant soil. By monitoring the hydrogen ion activity, which is represented by the potential for hydrogen, or PH, they operate. On a PH scale of 0 to 14, 0 represents severe acidity, 7 neutrality, and 14 alkalinity [?]. And in the sensor, its measuring range is between 3 and 9 PH, and its resolution is 0.1 PH [?].

Figure 3.4: Soil sensor

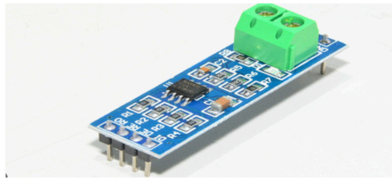


Figure 3.5: LC-electronics

3.4.5 Lithium Battery

It is one of the most important types of batteries, made of lithium compounds. It has several advantages, including its rechargeability, light weight, low cost, and safety. It is used in everyday electronics, from smartphones and laptops to electric vehicles [?]. Its operating principle is based on the storage and release of energy by moving lithium ions between the nodes during the charging and discharging process [?].

3.5 Our System Connection

The system architecture consists of many sensors that connect to an ESP32 microcontroller, which serves as the central processing unit and communication hub for all connected sensors. These sensors include DHT11, LDR, and an RS485-based soil sensor, which measure temperature, humidity, pH, and EC. A TTL to RS485 converter module is used to connect the soil sensor to the ESP32 using serial communication. Figure 3.6 shows how these components are connected.

The system components transfer information using digital and serial data interfaces. The LDR and DHT11 sensors connect to the ESP32 through GPIO digital pins to automatically transfer real-time light conditions, temperature, and humidity measurements from their sensors directly to the ESP32. The communication for the RS485 soil sensor depends on the TTL to RS485 converter to establish serial data exchange through the RS485 protocol. This setup makes it possible for the soil sensor and the ESP32 to communicate effectively over long distances, ensuring accurate and synchronized data collection from all connected sensors.

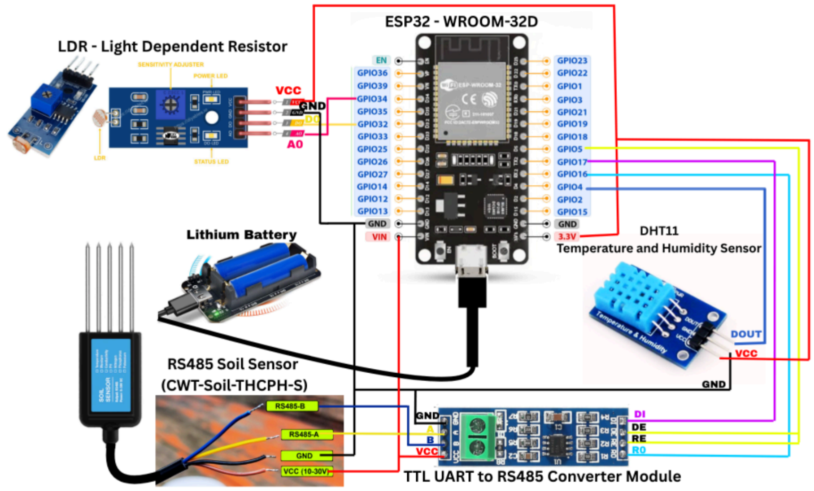


Figure 3.6: System connection.

3.6 Telegram Bot as a Communication Tool

We selected Telegram Bot as our communication platform because its real-time features ensure farmers receive immediate updates through their phones for plant health reports and sensor output from integrated AI disease detection and live sensor data monitoring. Also, users can access the interface through a familiar chat interface without needing to install complicated applications. In addition, its easy-to-use design is combined with secure, scalable operations and support for multiple languages. The combination of features like Google Colab integration and Arduino-based hardware capabilities made this system an optimal solution for connecting AI with agricultural field requirements.

3.7 System Architecture

After connecting all sensors to the ESP32 microcontroller, as outlined in our system connection section, we developed an integrated Arduino program to collect real-time environmental data from all sensors. The sensors include an analog LDR, a digital DHT11 humidity of air, and RS485 soil sensors that measure soil temperature, humidity, EC, and pH. The collected data is formatted as JSON and automatically uploaded every 30 seconds using connectivity to the specific JSON file on Google Drive using OAuth2 authentication and the Google Drive API in secure mode, with access managed through a refresh token to ensure continuous updates. The JSON file is overwritten with each new upload, maintaining the latest sensor readings.

In parallel, we implemented a Telegram bot using Google Colab that integrates AI-based plant disease detection. The bot analyzes leaf images through pre-trained YOLO models that reside on Google Drive to analyze user-submitted leaf images and detect leaf diseases that affect the plant. Additionally, the bot retrieves the latest environmental sensor data from the Google Drive JSON file and includes this information in its responses or sends it separately when users require it.

3.8 Conclusion

In conclusion, this chapter discusses the main steps and components used in the development of our system. We compare different IoT communication protocols and microcontrollers that we could use in our system and select the WIFI as a communication protocol and the ESP32 as a microcontroller which is the most appropriate choices. Also, the basic components of the system were presented in detail, which included the ESP32 microcontroller and sensors (DHT11, LDR, RS485) and how to connect them to form a fully functional system. In addition, the system provides real-time sensor information through a Telegram bot together with instant system alert capabilities. This section explains how every system element collaborates to fulfill the project aims through a detailed presentation of the system architecture. This approach provides a solid foundation for creating reliable and effective IoT solutions.

Chapter 4

Results

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4.1 Datasets with AI Training Results

4.1.1 Tomato

4.1.1.1 Tomato Dataset

The dataset for detecting tomato diseases was obtained from Mendeley Data and PlantVillage. Includes 11,172 images divided into three folders.

- Training set: 7,771 images (70%)
- Validation set: 2,307 images (20%)
- Test set: 1,094 images (10%)

The dataset includes three subsets, each containing eight classes: seven representing different tomato diseases (Gray Spot, Powdery Mildew, Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot) and one class for healthy plants (Tomato Healthy).

4.1.1.2 Tomato Training AI results

During training, the model operated for 77 epochs until early stopping criteria were applied when no changes happened to the performance metrics across 10 consecutive epochs. The model achieved its best output results at epoch 67.

Evaluation results showed strong performance: mAP of 99.46%, precision of 99.46%, and recall of 99.70%. Precision measures how many true positive outcomes the model generates from all its positive predictions, thus demonstrating its capability to reduce false positive results, while recall shows its effectiveness in detecting actual positives. The mAP metric gives a full overview of the detection accuracy of the model by measuring average precision over all classes and thresholds. These high scores across all categories and confidence levels indicate the model's ability to correctly identify nearly all positive cases and minimize missed detections.

The model training process shows different indicators for training and validation across epochs, as shown in Figure 4.1. The training losses presented in the first displays steadily decreasing box loss (train/box_loss) and classification loss (train/cls_loss), and distribution focal loss (train/dfl_loss,) which signals successful model learning and convergence. Precision and recall metrics (metrics/precision(B) ,metrics/recall(B)) improves continuously until both reach near-perfect values (≈ 1.0), reflecting that the model is becoming highly efficient at detecting true positives while minimizing false detections. The second row presents validation losses (val/box_loss, val/cls_loss, val/dfl_loss) which mostly decrease, with some variation caused by dataset complexity. Detection and localization performance demonstrate strong improvement during training, which results in mAP50(B) and mAP50-95(B) curves approaching 1.0. supported by smooth curves that demonstrate the stable and positive performance trends throughout the training epochs.

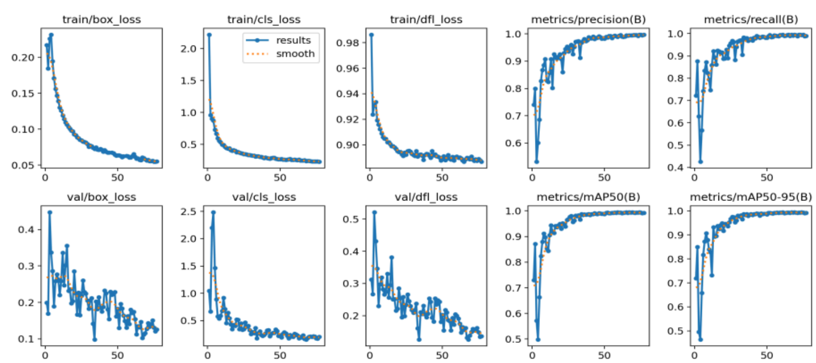


Figure 4.1: Different metrics with epochs during training and validation

A confusion matrix helps evaluate the performance of a classification model in ML by comparing the predicted values with the actual values of a dataset. The normalized confusion matrix for the tomato classification model, based on the test set in Figure 4.2, shows that most classes reached close to perfect accuracy and performance. Gray Spot, Early Blight, and Late Blight recorded slightly lower but still high accuracy (≈ 0.99), while Bacterial Spot, Healthy, and Leaf Mold reached perfect accuracy (1.00).

Small percentages of Gray Spot cases were misclassified as Powdery Mildew, and there is little confusion between Septoria Leaf Spot and nearby classes. The background class contains detections in which no valid tomato leaf was present, as well as noise images that do not belong to any class and hence cannot be classified; its low occurrence further highlights the model's robustness. Overall, the confusion matrix demonstrates that the model performs with high accuracy and efficiency.

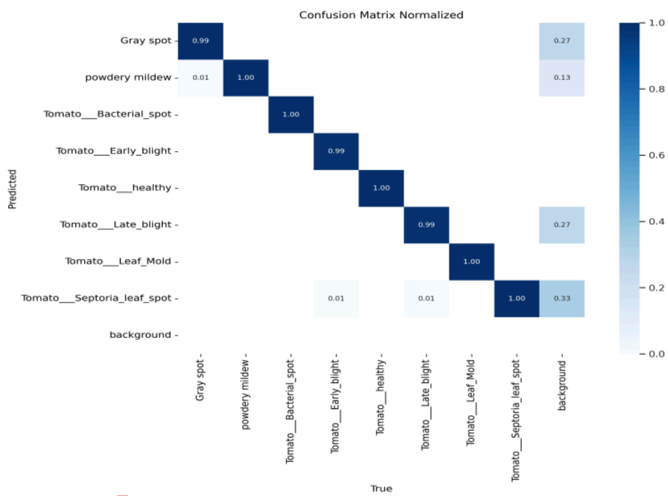


Figure 4.2: Confusion matrix normalized for tomato

Figure 4.3 shows the training and validation box loss curves. Training loss decreases steadily for effective learning, while validation loss has occasional changes but shows continuous gradual decline, indicating proper generalization capabilities at different epochs. Figure 4.4 displays the model's mAP at IoU thresholds 0.5 and 0.5:0.95. The model shows strong performance in disease detection and localization because its mAP values demonstrate a rapid increase and are stable above 0.95 after approximately 30 epochs. Figure 4.5 shows the precision and recall curves across training epochs. The model achieves a high rate of correct positive predictions with few false positives and false negatives, as shown by both metrics improving rapidly during early training and steady convergence towards values near 1.0. Overall, these figures show how well the model learns from the training data, performs well in localization and classification, and maintains a good balance between recall and precision, ultimately confirming the strength and robustness of the developed detection system.

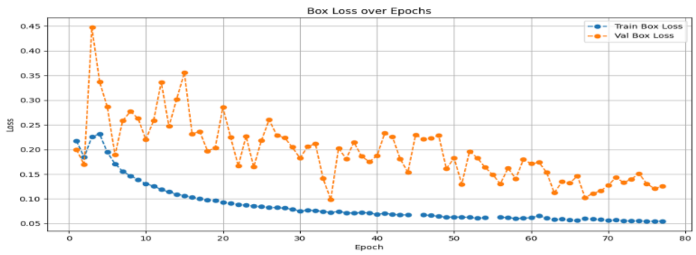


Figure 4.3: Training and validation box loss over epochs for tomato.

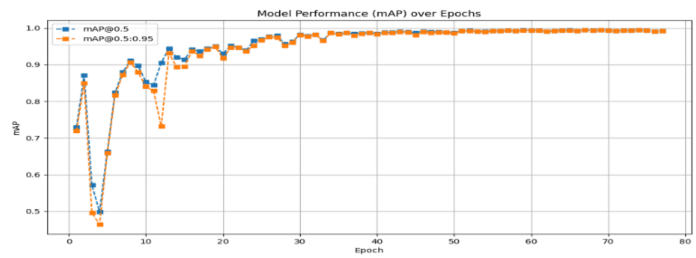


Figure 4.4: Model performance (mAP) over epochs for tomato

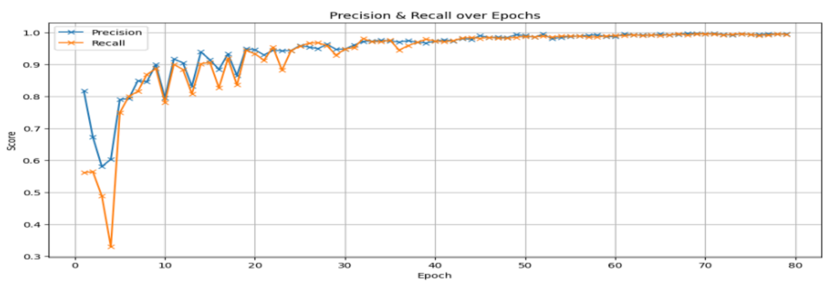


Figure 4.5: Precision-recall curve over epochs for tomato

Figure 4.6 shows four evaluation plots: precision-confidence, F1-confidence, precision-recall, and recall-confidence. The Precision-Confidence curve (top left) shows high precision across all confidence thresholds, particularly near 1.0, indicating excellent reliability and minimal false positives even as the confidence varies. The F1-Confidence curve (top right), which is the harmonic mean of precision and recall, changes with varying confidence thresholds. A constant and high F1 score over all ranges indicates a strong balance between precision and recall, resulting in reliable and consistent classification performance. The precision-recall curve (bottom left) displays the trade-off between precision and recall at different classification thresholds. The precision-recall curve shows closely clustered values across all classes, with a mAP@0.5 of 0.995, indicating an excellent balance of true positive detection and false positive reduction. Finally, the Recall-Confidence curve (bottom right) shows strong recall across all confidence levels, with just a little predicted reduction as confidence approaches 1.0. This indicates the model's ability to recognize true positive cases even at at stricter thresholds. Notably, for all classes, the model obtains an accuracy of 0.99 at a confidence threshold of 0.504 and an overall mAP of 0.995, demonstrating its excellent multi-class detection ability across several tomato disease categories and healthy samples.

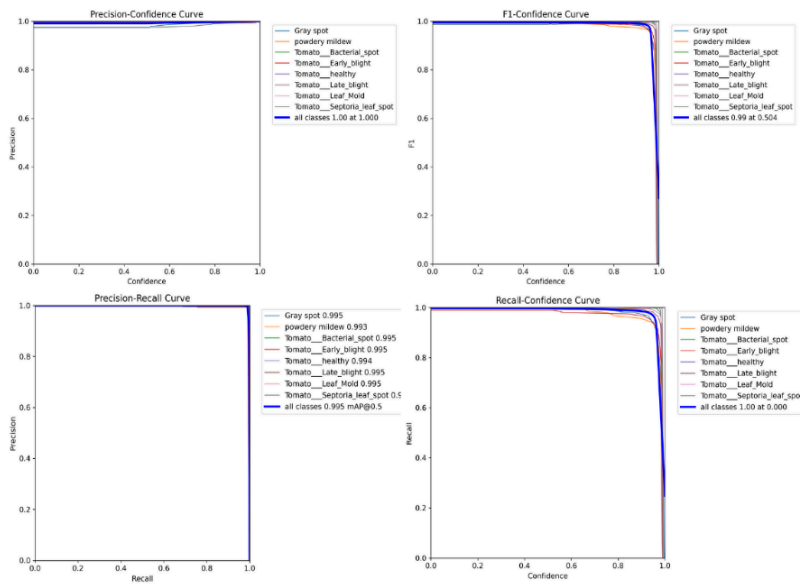


Figure 4.6: Model performance curves for tomato disease detection.

4.1.2 Lettuce

4.1.2.1 Lettuce Dataset

The dataset selected for detecting lettuce diseases was obtained from Roboflow and Kaggle. Includes 8451 images divided into three folders.

- Training set: 6,288 images (70%)
- Validation set: 1,147 images (20%)
- Test set: 1,016 images (10%)

The dataset comprises three subsets, each containing seven classes: six presenting different lettuce diseases (Bacterial, Downy Mildew, Mosaic Virus, Powdery Mildew, Septoria Blight, Fusarium wilt) and one presenting healthy lettuce plants (Healthy Lettuce).

4.1.2.2 Lettuce AI Training results

During training, the model operated for 114 epochs until early stopping criteria were applied, so no improvement is shown in the last 20 epochs. Best results observed at epoch 94.

The mAP of the training is 98.07%, which shows a strong overall object detection capability, precision: 90.90% shows a high percentage positive predictive value, and recall: 93.59%, which also shows a high percentage of identifies positive instances (true positives). In conclusion, from these results the model indicate a high accuracy with very small false positives and false negatives.

Figure 4.7 shows how a model trains and validates through 114 epochs by demonstrating both performance and loss components. The loss curves exhibit a constant downward trend, indicating that the model learns efficiently throughout its training process. The continuous improvement of precision and recall metrics until they reach almost perfect levels close to 1, indicating better model accuracy along with improved prediction completeness. While the mAP50 and mAP50-95 demonstrate steady improvement as the model gains enhanced performance at various thresholds of intersection-over-union. The results show strong training performance along with low signs of overfitting, which matches the regular alignment of training and validation curves.

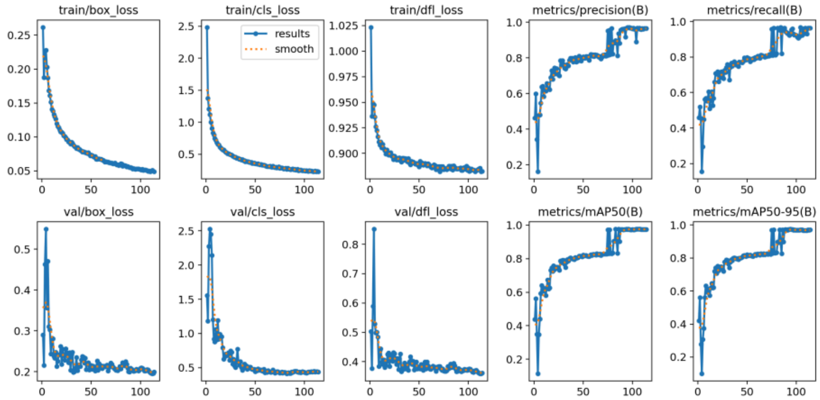


Figure 4.7: Different metrics with epochs during training and validation

Figure 4.8 shows the normalization of a confusion matrix of the lettuce, which analyzes the model's classification performance between seven categories, including the six lettuce diseases, healthy plants, and background. The diagonal dominance shows values close to 1.0, indicating a strong ability of the model to correctly classify most classes, such as Wilt and Leaf Blight and healthy both of which achieve perfect or near-perfect prediction rates. Minor classification errors were observed; for example, downy mildew on lettuce was sometimes confused with other diseases at low rates. The off-diagonal values, especially for the background and healthy classes, show small but noticeable errors. Overall, the matrix demonstrates high classification accuracy, with very few significant confusions between classes, demonstrating the model's robustness in distinguishing between different lettuce disease types and health conditions.

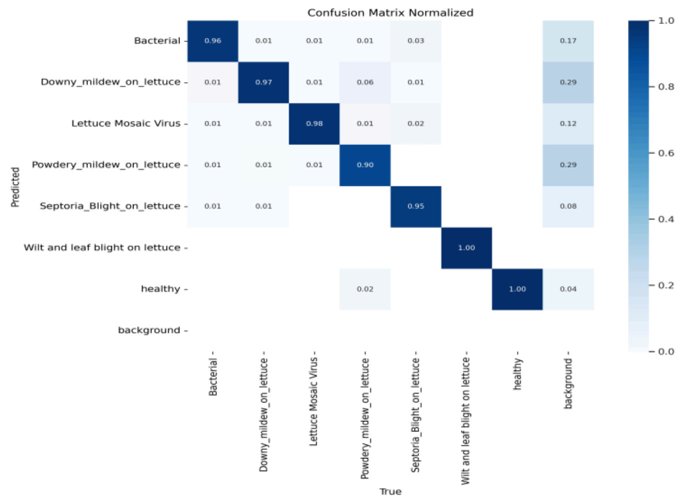


Figure 4.8: Confusion matrix normalized for lettuce

The three figures below demonstrate the model performance during training and evaluation in lettuce disease detection procedures. Figure 4.9 illustrates box loss progress during training epochs as both validation and training values decrease progressively, indicating effective learning. However, in the later epochs, the slight gap between training and validation losses suggests minor overfitting. The model shows improving mAP values through epochs, as Figure 4.10 shows strong detection ability using both mAP@0.5 and mAP@0.5:0.95 metrics, which reach values approaching 1.0. Figure 4.11 shows how precision-recall curves evolve across epochs; they both show consistent rise and stable growth rates with very positive rates. The combined results demonstrate that the model has successful, robust training with effective generalization, resulting in high accuracy in detecting lettuce diseases.

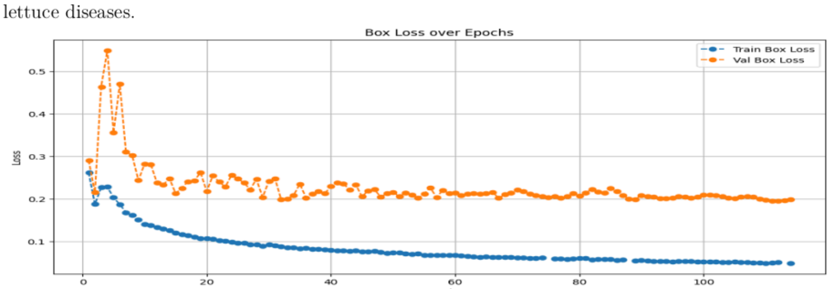


Figure 4.9: Training and validation box loss over epochs for lettuce

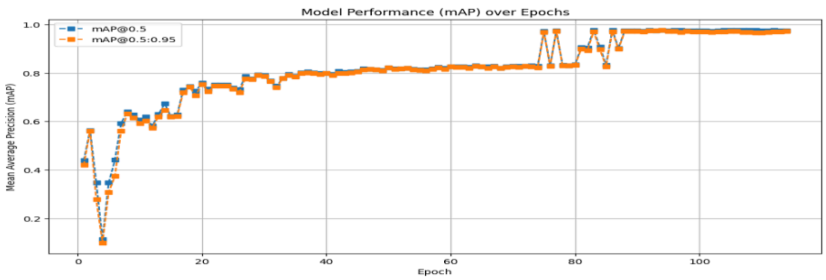


Figure 4.10: Model performance (mAP) over epochs for lettuce

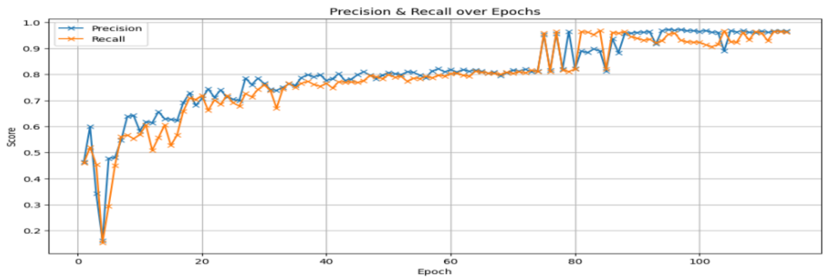


Figure 4.11: Precision-Recall curve over epochs for lettuce

Figure 4.12 presents four evaluation curves to evaluate the performance of a multi-class classification detection model that identifies lettuce conditions, including diseases and health statuses. The fist and last figures display precision and recall curves, which shows how they are vary through different confidence thresholds across individual classes, with the precision achieving a perfect score of 1.00 at 1.000 for all classes, while the overall recall reflects consistent values of 0.97 at a confidence of 0.900. The F1-Confidence Curve demonstrates stable performance because it indicates how well the F1 score balances precision against recall when system confidence reaches higher levels with overall F1 score of 0.93 at confidence of 0.904 across all classes. The precision-recall curve presented in the bottom left section shows that the system maintains high performance levels across most classes with minimal drop-off, where the macro-averaged performance yields a strong (mAP) of 0.981 at IoU=0.5.

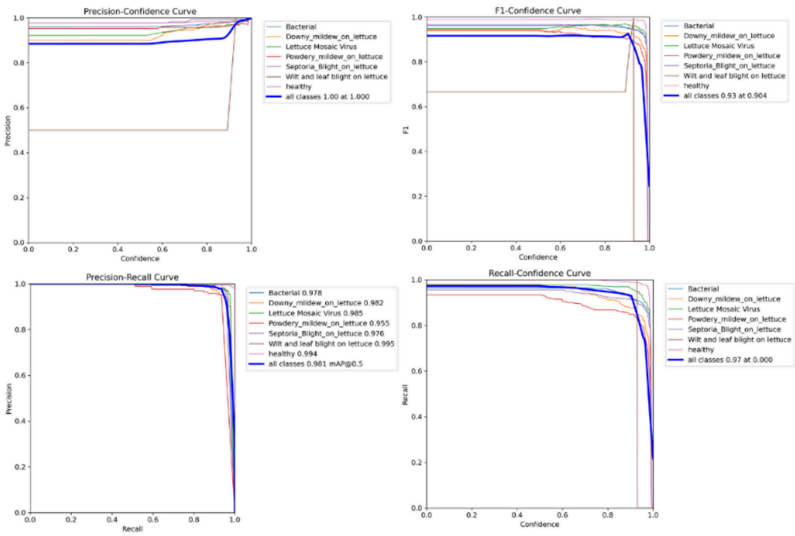


Figure 4.12: Model performance curves for lettuce disease detection.

4.1.3 Cucumber

4.1.3.1 Cucumber Dataset

The dataset selected for detecting cucumber diseases was obtained from Roboflow and Kaggle. Includes 5,453 images divided into three subsets.

- Training set: 3,251 images (60%)
- Validation set: 1,050 images (20%)
- Test set: 1,152 images (20%)

The dataset includes three subsets, each containing six classes: five representing different cucumber diseases (Anthracnose, Bacterial Wilt, Downy Mildew, Gummy Stem Blight, and Powdery Mildew) and one representing healthy cucumber plants (Fresh Leaf).

4.1.3.2 Cucumber Training AI results

During training, the model operated for 59 epochs until early stopping training criteria were applied. There was no improvement noticed during the last 20 epochs. Epoch 39 gave the best results.

Tht mAP of the training is 99.05%, demonstrating strong potential for detecting all objects. Precision is 98.36%, demonstrating high positive results, and the recall is 97.71%. The model demonstrates high capability to detect a large number of actual positive cases.

Figure 4.13 illustrates the ways a model is trained and validated using 39 epochs by demonstrate performance and loss components. The loss curves show a consistent downward trend, reflecting efficient model learning. Precision and recall steadily improve by approaching a value of 1, approaching near-perfect levels, while mAP@0.5 and mAP@0.5:0.95 increase steadily, indicating strong performance across IoU thresholds. The close alignment of training and validation curves suggests effective training with minimal overfitting.

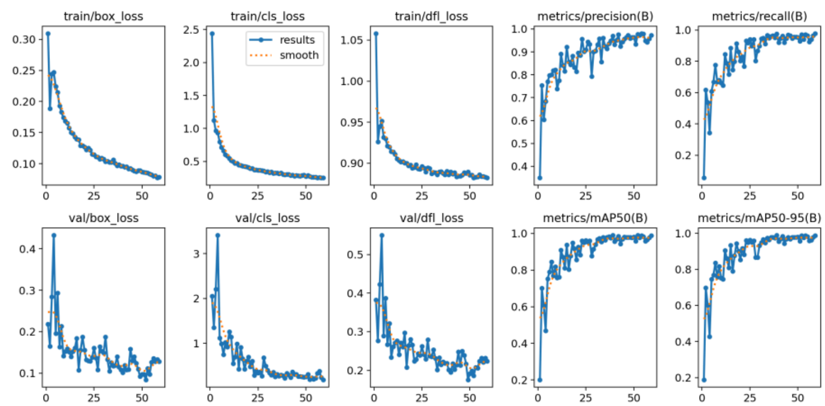


Figure 4.13: Different metrics with epochs during training and validation

Figure 4.14 presents the normalized confusion matrix, which analyzes the model's classification performance between seven categories, including the six cucumber diseases, fresh plants, and background. Most classification categories in this model receive accurate predictions because strong correlations exist in the diagonal section, where Bacterial Wilt and Fresh Leaf demonstrate either perfect or near-perfect results (up to 1.0).

Some minor misclassifications are present. Approximately 0.05 percent of downy mildew classifications are misidentified as anthracnose. The minimal prediction errors in the matrix are revealed through off-diagonal values because two categories with similar symptoms share these errors.

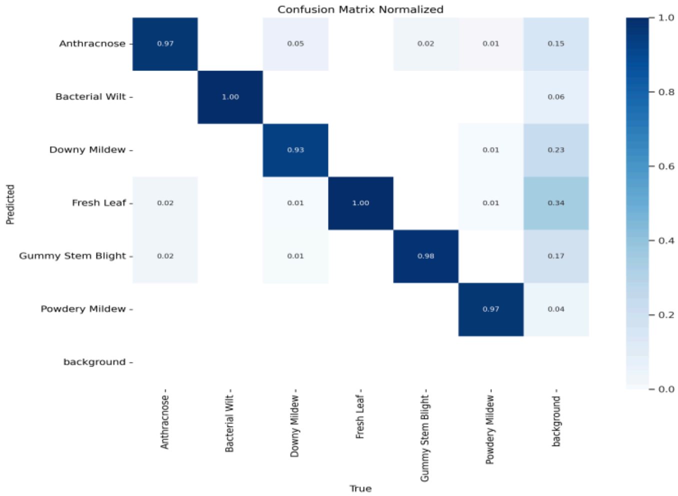


Figure 4.14: Confusion matrix normalized for cucumber

The Figures below show how the model functions during training and evaluation of cucumber disease detection procedures. A progressive drop in box loss values during the training process and validation stages demonstrates learning effectiveness according to Figure 4.15. The model presented indicates improved mAP values with epochs as can be seen in Figure 4.16 this being strong. detection capability upon mAP@0.5 and mAP@0.5:095, which get close to 1.0. Figure 4.17 shows the precision-recall curves by epochs as they change. both of them have a constant increase and steady growth rates with very optimistic rates. The aggregate results show that the model has a robust and successful training and generalization, leading to high accuracy in detecting cucumber diseases.

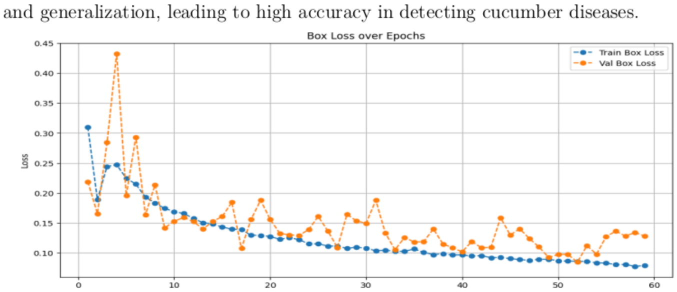


Figure 4.15: Cucumber box loss over epochs.

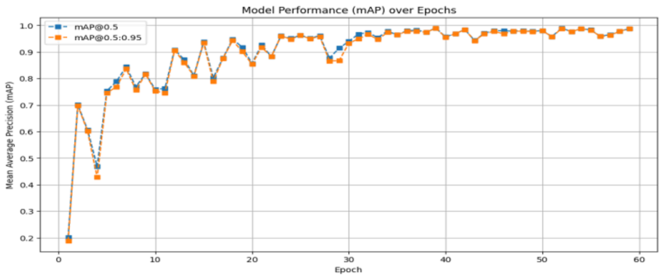


Figure 4.16: Cucumber model performance over epochs.

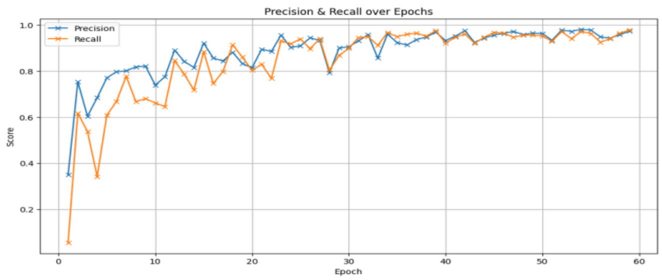


Figure 4.17: Cucumber precision-recall curves over epochs

The performance evaluation of multi-class classification appears in Figure 4.18. The designed model demonstrates effective performance by identifying both health and various disease states present in cucumber leaves. The model classifies all objects to a precision of 1.0 at a confidence level of 1.0, indicating perfect classification accuracy even at the highest confidence levels. The sensitivity of the classes remains reliable between 0.97 and 1.00 for all recall values above the confidence threshold of 0.9. The tested system demonstrates robust performance through F1 scores, which stay strong at every confidence threshold and reach their maximum of 0.98 at the confidence level 0.839. The Precision-Recall Curve demonstrates excellent clustering of precision-recall relationships among all classes because it shows limited drops, which confirms stable detection performance under differing threshold values. The model demonstrates strong generalization ability and robust accuracy in multi-class cucumber disease detection.

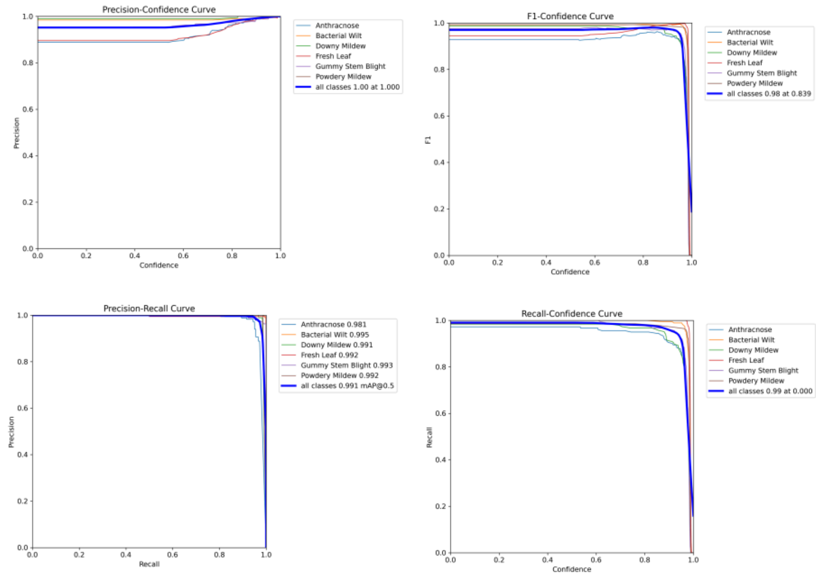


Figure 4.18: Model performance curves for cucumber disease detection.

4.2 Summary of AI Model Training Results

The training process displayed different numbers of epochs and early stopping points during the process. These changes were mainly due to factors like the size and complexity of the dataset, the speed and capacity of the GPU, and internet stability, especially when using cloud platforms like Google Colab. Other system factors, such as available memory and processing workload, also affected the training performance.

Our performance metrics for applying the YOLOv11 model on lettuce, cucumber, and tomato plant disease datasets appear in Table 4.1. The table presents information about the image, class numbers, shows accuracy measurements, precision, and recall statistics for the training, validation, and testing phases of each crop. The model demonstrates stable performance across all datasets by achieving high accuracy rates and strong detection capabilities.

Table 4.1: Performance Metrics for Lettuce, Cucumber, and Tomato Datasets Using YOLOv11.

Plant	No. of Images	No. of Classes	Training Accuracy	Validation Accuracy	Testing Accuracy	Precision	Recall
Lettuce	8,451	7	98.07%	94.69%	96.95%	90.09%	93.59%
Cucumber	5,453	6	99.05%	96.00%	88.63%	98.36%	97.91%
Tomato	11,172	8	99.45%	96.49%	96.71%	99.12%	99.80%

4.3 Compare Our AI Training Results with Similar Work

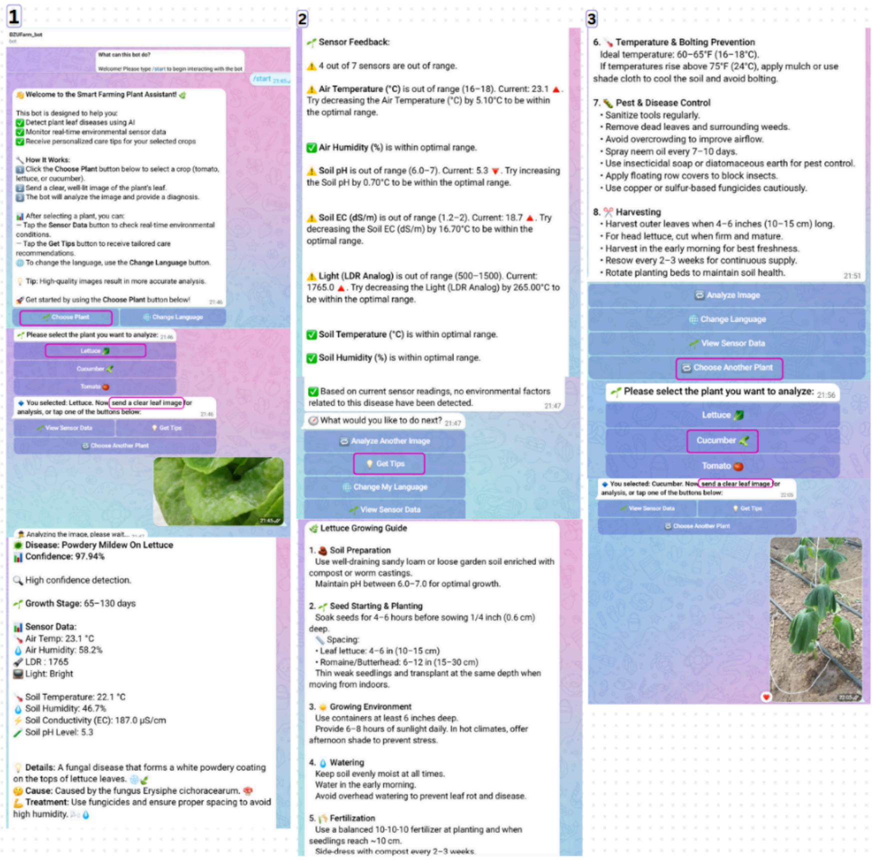
Table 4.1 shows our performance result of plant disease detection using the YOLOv11 model for lettuce, cucumber, and tomato. Compared to similar work that was published previously, as shown in Table 4.2, our model achieved higher or similar performance on most evaluation metrics. For example, for the tomato, our model demonstrated a training accuracy of 99.45% and a test accuracy of 96.71%. However, other models have similar test accuracies (e.g., 97.96% and 98.75%) but using smaller datasets with fewer classes and fewer images. For lettuce, our technique attained a test accuracy of 96.95%, which exceeded the previous research models with 81% and 82.28% accuracy rates. Also, our lettuce dataset contained 8,451 images with 7 classes, which provided better generalization than previously used 3-class datasets with smaller datasets. The train accuracy of cucumber reported by a previous CNN model at 99.95% was exceeded by our YOLOv11, which worked with a dataset featuring 6 classes and 5,453 images, beyond previous models that used only 3 classes. The large, diverse dataset combined with the YOLOv11 detection model gives our approach superior accuracy along with better generalization capabilities across different plant types and precision-recall performance.

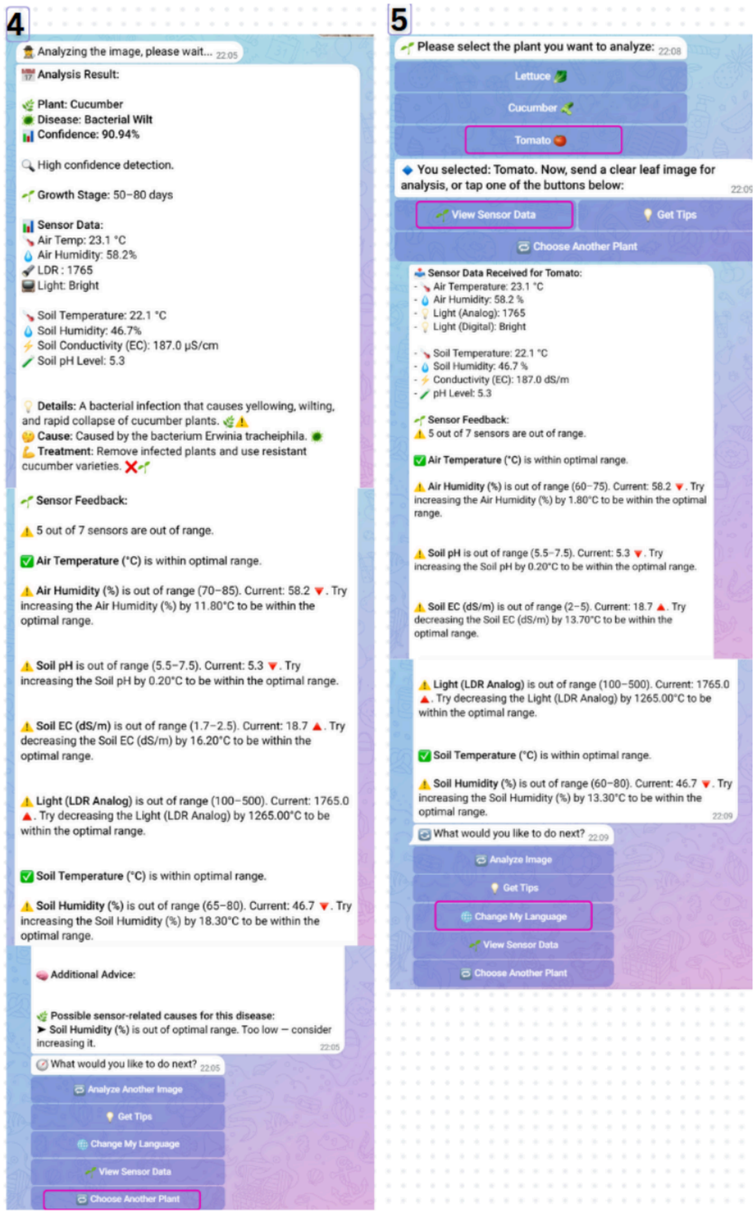
Table 4.2: Similar Work of Plant Disease Detection Models

Plant	Model Used	Dataset Name	No. of Images	No. of Classes	Train Accuracy	Test Accuracy
Tomato	CNN	Tomato Leaf Disease [?]	11,000	10	97.50%	95.00%
Tomato	CNN	Tomato Leaf Disease [?]	3,600	3	99.60%	98.75%
Tomato	CNN	Tomato Leaf Disease Prediction [?]	7,579	5	98.48%	97.96%
Cucumber	CNN	Cucumber Leaf Disease Classification [?]	3,754	3	99.95%	94.00%
Cucumber	VGG16	Cucumber Leaf Disease Classification [?]	3,754	3	98.93%	96.00%
Lettuce	CNN	Lettuce Plant Disease [?]	2,813	3	84.75%	82.28%
Lettuce	Roboflow 3.0 (Custom Object Detection - Fast)	Lettuce Disease Detection [?]	10,339	8	77.80%	81.00%

4.4 Telegram Bot Results

BZU FarmBot is the Telegram bot that connects the farmer with his farm. Through it, the farmer sends a clear photo of the plant leaf that may be infected with the disease, and the bot sends the disease name. Also, the farmer can see the sensor value and get some tips. First the farmer open the bot through a link then the bot send a start message to start the conversation as shown in Figure 4.19. After that, a select language message occur to choose Arabic or English. Then, depending on the language, the bot start with a welcome message that describe the bot purpose in short with a button to choose the plant that want to analyze between the three plants: lettuce, cucumber, and tomato then depend on the selection plant the bot send four choices for the farmer to send an image to analyze it and give the disease name and a small description about it like how the symptoms are shown in the plant and what causes it, with the sensor values, and see if any of the sensor values cause this disease. Or immediately display the sensor values and check if any of them is not in the optimal range and warn the farmer about it. Or get some tips about the plant, which includes a detailed explanation of how to grow the plant and how to avoid diseases and pests, and the last button choose another plant to analyze. With each choice from the four choices, there are five buttons shown that do the same thing: analyze another image for the same plant, get tips about the plant, change the language, view the sensor data for the plant, or even change the plant to repeat the same thing.


1
Welcome to the Smart Farming Plant Assistant!
This bot is designed to help you:
Detect plant leaf diseases using AI
Monitor real-time environmental sensor data
Receive personalized care tips for your selected crops
How It Works:
Click the Choose Plant button below to select a crop (tomato, lettuce, or cucumber).
Send a clear, well-lit image of the plant's leaf.
The bot will analyze the image and provide a diagnosis.
After selecting a plant, you can:
– Tap the Sensor Data button to check real-time environmental conditions.
– Tap the Get Tips button to receive tailored care recommendations.
To change the language, use the Change Language button.
Tip: High-quality images result in more accurate analysis.
Get started by using the Choose Plant button below!
Choose Plant
Change Language
Please select the plant you want to analyze:
Lettuce
Cucumber
Tomato
You selected: Lettuce. Now send a clear leaf image for analysis, or tap one of the buttons below:
View Sensor Data
Get Tips
Choose Another Plant
Analyzing the image, please wait...
Disease: Powdery Mildew On Lettuce
Confidence: 97.94%
High confidence detection.
Growth Stage: 65–130 days
Sensor Data:
Air Temp: 23.1 °C
Air Humidity: 58.2%
LDR : 1765
Light: Bright
Soil Temperature: 22.1 °C
Soil Humidity: 46.7%
Soil Conductivity (EC): 187.0 µS/cm
Soil pH Level: 5.3
Details: A fungal disease that forms a white powdery coating on the tops of lettuce leaves.
Cause: Caused by the fungus Erysiphe cichoracearum.
Treatment: Use fungicides and ensure proper spacing to avoid high humidity.
2
Sensor Feedback:
4 out of 7 sensors are out of range.
Air Temperature (°C) is out of range (16–18). Current: 23.1. Try decreasing the Air Temperature (°C) by 5.10°C to be within the optimal range.
Air Humidity (%) is within optimal range.
Soil pH is out of range (6.0–7). Current: 5.3. Try increasing the Soil pH by 0.70°C to be within the optimal range.
Soil EC (dS/m) is out of range (1.2–2). Current: 18.7. Try decreasing the Soil EC (dS/m) by 16.70°C to be within the optimal range.
Light (LDR Analog) is out of range (500–1500). Current: 1765.0. Try decreasing the Light (LDR Analog) by 265.00°C to be within the optimal range.
Soil Temperature (°C) is within optimal range.
Soil Humidity (%) is within optimal range.
Based on current sensor readings, no environmental factors related to this disease have been detected.
What would you like to do next?
Analyze Another Image
Get Tips
Change My Language
View Sensor Data
Lettuce Growing Guide
1. Soil Preparation
Use well-draining sandy loam or loose garden soil enriched with compost or worm castings.
Maintain pH between 6.0–7.0 for optimal growth.
2. Seed Starting & Planting
Soak seeds for 4–6 hours before sowing 1/4 inch (0.6 cm) deep.
Spacing:
• Leaf lettuce: 4–6 in (10–15 cm)
• Romaine/Butterhead: 6–12 in (15–30 cm)
Thin weak seedlings and transplant at the same depth when moving from indoors.
3. Growing Environment
Use containers at least 6 inches deep.
Provide 6–8 hours of sunlight daily. In hot climates, offer afternoon shade to prevent stress.
4. Watering
Keep soil evenly moist at all times.
Water in the early morning.
Avoid overhead watering to prevent leaf rot and disease.
5. Fertilization
Use a balanced 10-10-10 fertilizer at planting and when seedlings reach ~10 cm.
Side-dress with compost every 2–3 weeks.
3
6. Temperature & Bolting Prevention
Ideal temperature: 60–65°F (16–18°C).
If temperatures rise above 75°F (24°C), apply mulch or use shade cloth to cool the soil and avoid bolting.
7. Pest & Disease Control
• Sanitize tools regularly.
• Remove dead leaves and surrounding weeds.
• Avoid overcrowding to improve airflow.
• Spray neem oil every 7–10 days.
• Use insecticidal soap or diatomaceous earth for pest control.
• Apply floating row covers to block insects.
• Use copper or sulfur-based fungicides cautiously.
8. Harvesting
• Harvest outer leaves when 4–6 inches (10–15 cm) long.
• For head lettuce, cut when firm and mature.
• Harvest in the early morning for best freshness.
• Resow every 2–3 weeks for continuous supply.
• Rotate planting beds to maintain soil health.
Analyze Image
Change Language
View Sensor Data
Choose Another Plant
Please select the plant you want to analyze:
Lettuce
Cucumber
Tomato
You selected: Cucumber. Now send a clear leaf image for analysis, or tap one of the buttons below:
View Sensor Data
Get Tips
Choose Another Plant


4
Analyzing the image, please wait... 22:05
Analysis Result:
Plant: Cucumber
Disease: Bacterial Wilt
Confidence: 90.94%
High confidence detection.
Growth Stage: 50–80 days
Sensor Data:
Air Temp: 23.1 °C
Air Humidity: 58.2%
LDR : 1765
Light: Bright
Soil Temperature: 22.1 °C
Soil Humidity: 46.7%
Soil Conductivity (EC): 187.0 µS/cm
Soil pH Level: 5.3
Details: A bacterial infection that causes yellowing, wilting, and rapid collapse of cucumber plants.
Cause: Caused by the bacterium Erwinia tracheiphila.
Treatment: Remove infected plants and use resistant cucumber varieties.
Sensor Feedback:
5 out of 7 sensors are out of range.
Air Temperature (°C) is within optimal range.
Air Humidity (%) is out of range (70–85). Current: 58.2. Try increasing the Air Humidity (%) by 11.80°C to be within the optimal range.
Soil pH is out of range (5.5–7.5). Current: 5.3. Try increasing the Soil pH by 0.20°C to be within the optimal range.
Soil EC (dS/m) is out of range (1.7–2.5). Current: 18.7. Try decreasing the Soil EC (dS/m) by 16.20°C to be within the optimal range.
Light (LDR Analog) is out of range (100–500). Current: 1765.0. Try decreasing the Light (LDR Analog) by 1265.00°C to be within the optimal range.
Soil Temperature (°C) is within optimal range.
Soil Humidity (%) is out of range (65–80). Current: 46.7. Try increasing the Soil Humidity (%) by 18.30°C to be within the optimal range.
Additional Advice:
Possible sensor-related causes for this disease:
Soil Humidity (%) is out of optimal range. Too low — consider increasing it.
22:05
What would you like to do next? 22:05
Analyze Another Image
Get Tips
Change My Language
View Sensor Data
Choose Another Plant
5
Please select the plant you want to analyze: 22:08
Lettuce
Cucumber
Tomato
You selected: Tomato. Now, send a clear leaf image for analysis, or tap one of the buttons below:
22:09
View Sensor Data
Get Tips
Choose Another Plant
Sensor Data Received for Tomato:
- Air Temperature: 23.1 °C
- Air Humidity: 58.2 %
- Light (Analog): 1765
- Light (Digital): Bright
- Soil Temperature: 22.1 °C
- Soil Humidity: 46.7 %
- Conductivity (EC): 187.0 dS/m
- pH Level: 5.3
Sensor Feedback:
5 out of 7 sensors are out of range.
Air Temperature (°C) is within optimal range.
Air Humidity (%) is out of range (60–75). Current: 58.2. Try increasing the Air Humidity (%) by 1.80°C to be within the optimal range.
Soil pH is out of range (5.5–7.5). Current: 5.3. Try increasing the Soil pH by 0.20°C to be within the optimal range.
Soil EC (dS/m) is out of range (2–5). Current: 18.7. Try decreasing the Soil EC (dS/m) by 13.70°C to be within the optimal range.
Light (LDR Analog) is out of range (100–500). Current: 1765.0. Try decreasing the Light (LDR Analog) by 1265.00°C to be within the optimal range.
Soil Temperature (°C) is within optimal range.
Soil Humidity (%) is out of range (60–80). Current: 46.7. Try increasing the Soil Humidity (%) by 13.30°C to be within the optimal range.
22:09
What would you like to do next? 22:09
Analyze Image
Get Tips
Change My Language
View Sensor Data
Choose Another Plant


6
Please choose your language:
يرجى اختيار لغتك:
English
العربية
تم اختيار اللغة العربية.
اضغط على الزر أدناه للبدء!
ابدأ الزراعة الذكية
مرحبًا بك في مساعد الزراعة الذكية!
يساعدك هذا البوت في:
كشف أمراض أوراق النباتات باستخدام الذكاء الاصطناعي
مراقبة بيانات الحساسات البيئية في الوقت الفعلي
الحصول على نصائح رعاية مخصصة للمحاصيل التي تختارها
طريقة الاستخدام:
1 اضغط على زر اختيار المحصول أدناه لاختيار نوع المحصول (طماطم، خس، أو خيار).
2 أرسل صورة واضحة ومضاءة جيدًا لورقة النبات.
3 سيقوم البوت بتحليل الصورة وتقديم التشخيص المناسب.
بعد اختيار المحصول، يمكنك:
– الضغط على زر بيانات الحساسات للاطلاع على القراءات البيئية الفعلية.
– الضغط على زر الحصول على نصائح للحصول على توصيات رعاية مفصلة.
لتغيير اللغة، استخدم زر تغيير اللغة.
نصيحة: جودة الصورة العالية تؤدي إلى نتائج تحليل أكثر دقة.
ابدأ الآن باستخدام زر اختر المحصول أدناه!
اختر المحصول
تغيير اللغة
اختر نوع النبات الذي ترغب في تحليله:
خس
خيار
طماطم
لقد اخترت: طماطم . الآن، أرسل صورة واضحة للورقة لإجراء التحليل،
أو اضغط على أحد الأزرار أدناه:
عرض بيانات الحساسات
الحصول على نصائح
اختيار نبتة أخرى
جارٍ تحليل الصورة، يرجى الانتظار...


7
نتيجة التحليل:
المحصول: طماطم
المرض: بقعة الأوراق السبتورية
Tomato Septoria Leaf Spot
الدقة: 99.28%
اكتشاف بدرجة ثقة عالية
مرحلة النمو: 60–85 أيام
بيانات الحساسات:
درجة حرارة الهواء: 23.1 °C
رطوبة الهواء: 58.2%
قيمة الضوء: 1765
شدة الضوء: مشرق
درجة حرارة التربة: 22.1 °C
رطوبة التربة: 46.7%
التوصيلية الكهربائية (EC): 187.0 µS/cm
مستوى الحموضة (pH): 5.3
التفاصيل: مرض فطري يسبب بقع داكنة مع هالة صفراء على أوراق الطماطم.
السبب: تسببه الفطريات Septoria lycopersici.
العلاج: استخدم المبيدات الفطرية وأزل الأوراق المصابة.
نتيجة الحساسات:
5 من أصل 7 من الحساسات خارج النطاق.
درجة حرارة الهواء (س°) ضمن النطاق المثالي.
رطوبة الهواء (%) خارج النطاق (60–75). الحالي: 58.2 ▼. حاول زيادة رطوبة الهواء (%) بمقدار 1.80س° ليكون ضمن النطاق المثالي.
مستوى الحموضة (pH) خارج النطاق (5.5–7.5). الحالي: 5.3 ▼. حاول زيادة مستوى الحموضة (pH) بمقدار 0.20س° ليكون ضمن النطاق المثالي.
التوصيلية الكهربائية (EC) خارج النطاق (2–5). الحالي: 18.7 ▲. حاول تقليل التوصيلية الكهربائية (EC) بمقدار 13.70 س° ليكون ضمن النطاق المثالي.
قيمة الضوء خارج النطاق (100–500). الحالي: 1765.0 ▲. حاول تقليل قيمة الضوء بمقدار 1265.00 س° ليكون ضمن النطاق المثالي.
درجة حرارة التربة (س°) ضمن النطاق المثالي.
رطوبة التربة (%) خارج النطاق (60–80). الحالي: 46.7 ▼. حاول زيادة رطوبة التربة (%) بمقدار 13.30س° ليكون ضمن النطاق المثالي.
نصيحة إضافية:
الأسباب المحتملة المتعلقة بالحساسات لهذا المرض:
➤ درجة حموضة التربة (Soil pH) خارج النطاق المثالي. منخفض جدًا – يُفضل زيادته.
ماذا تريد أن تفعل بعد ذلك؟
تحليل صورة جديدة
الحصول على نصيحة
تغيير اللغة
عرض بيانات الحساس
اختيار نبتة أخرى

Figure 4.19: Results on telegram bot.

Chapter 5

Conclusion and Future Work

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5.1 Conclusion

In conclusion, the agriculture system is a vital part of improving Palestine's farming. But it faces significant challenges so innovative solutions are needed to support farmers and enhance productivity. This project aims to transform agriculture by integrating AI and sensor technologies to ensure high accuracy in diagnosing plant health, while soil and environmental data provide tailored farming practices. The project demonstrates the potential to improve the quality and productivity of essential crops like tomatoes, lettuce, and cucumbers by precise identification of plant diseases using AI models where it get a picture of the plant leaf suspected of being infected with the disease, analyze it, and provide the farmer with the result on Telegram. While the sensors work together to collect environmental data about the soil and then provide the farmer with it, DHT11 provides temperature and humidity air data, the LDR sensor gives light intensity to the plants, and the RS485 senso determines the temperature and humidity of the soil with the electrical conductivity of the soil and the PH value which measures acidity or alkalinity in the plant soil and then sends data through the ESP32 microcontroller. All these sensors work together with the AI model to provide information about the plant's health to the farmer through a Telegram Bot that allows them to remotely monitor their crops, track plant growth stages, and make informed decisions based on real-time data. This project emphasizes on the importance of smart agriculture in addressing food security challenges, supporting farmers, and ensuring a flexible agricultural sector.

5.2 Future Work

Our future work focuses on expanding the system to include large farm areas, where more than one ESP32 is connected via the ESP-Now protocol. Data will be transferred between these nodes and ultimately sent to a central receiving ESP32, which is connected to a LoRa module to transmit information to the user over long distances. Also, the AI model in the system should be extended to handle additional crop types and increase the number of diseases in each crop dataset. In addition to that, the system can be enhanced to make real-time video analysis by using a Raspberry Pi, for example, to give live monitoring for plants to instantly diagnose diseases without waiting for an image from the farmer. Furthermore, improve the YOLO model by applying the newest version, and also connect the Telegram bot with the cloud database. To save previous sensor readings, images, and disease detection results. So that the user can have a history of their information and analyze the change of this information over time.

Publication

As part of our project achievements, our research paper titled “Towards Smart Farming: An IoT-Based Optimized Agriculture System” has been accepted for presentation at the International Conference on Computer and Communication Engineering (ICCCE 2025) on Jun 4, to be held in Kuala Lumpur, Malaysia. The acceptance was granted subject to minor revisions based on reviewer feedback and adherence to the IEEE formatting guidelines. This recognition reflects the novelty and relevance of our work in the field of smart agriculture and reinforces the academic contribution of our project on a global platform.

Conference	Paper title (details)	Status
ICCCE 2025	Towards Smart Farming an IoT Based Optimized Agriculture System	Accepted

Figure 5.1: Accepted our paper in ICCCE2025.

Access here to see our paper:
https://drive.google.com/drive/folders/1dY_QNDR0VnpXAIknG26i6YFC1oCoedJE?usp=drive_link

Appendices

All essential project files are organized in this Google Drive folder, including:

Training codes: written in Colab notebooks that evaluate plant disease detection models for all crops, with saved results and model output data for each one.

Plant datasets: arranged and preprocessed for performing disease classification.

Arduino Code: for connecting environmental sensors like the soil sensor_RS485 with DHT11 and light sensors while working with the ESP32 using Arduino IDE.

Telegram Bot Integration code: Python-based code in Google Colab to create disease detection and sensor monitoring through AI.

Access here:

`https://drive.google.com/drive/folders/1pWgFdSLCarY_-aUb_4DLjbf6yl_GhCBT?usp=drive_link`

Or scan

Telegram Bot Link:

`https://t.me/BZUFarm_bot`

Or scan

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